

MAGNETOM

Safety-Related Tests Procedure System

Safety-Related Tests according to IEC 62353

Aera
Skyra

The protocol M7-010.834.05.03.XX is required for these instructions

Document Version

Siemens reserves the right to change its products and services at any time.

In addition, manuals are subject to change without notice. The hardcopy documents correspond to the version at the time of system delivery and/or printout. Versions to hardcopy documentation are not automatically distributed.

Please contact your local Siemens office to order a current version or refer to our website <http://www.healthcare.siemens.com>.

Disclaimer

Siemens provides this documentation “as is” without the assumption of any liability under any theory of law.

The content described herein requires superior understanding of our equipment and may only be performed by qualified personnel who are specially trained for such installation and/or service.

Copyright

“© Siemens, 2015” refers to the copyright of a Siemens entity such as:

Siemens Healthcare GmbH - Germany

Siemens Aktiengesellschaft - Germany

Siemens Shenzhen Magnetic Resonance Ltd. - China

Siemens Shanghai Medical Equipment Ltd. - China

Siemens Healthcare Private Ltd. - India

Siemens Medical Solutions USA Inc. - USA

Siemens Healthcare Diagnostics Inc. - USA and/or

Siemens Healthcare Diagnostics Products GmbH - Germany

1	General Information	5
1.1	Safety Information	5
1.2	General remarks	6
1.3	Documentation	7
1.4	Acronyms:	8
1.5	Time Frame for the Safety-Related Tests.	9
2	General Tests	11
2.1	System	11
2.2	Options and accessories	12
2.2.1	Tim Dockable Table (optional)	12
2.2.2	Combi Dockable Table (optional)	12
2.2.3	Surface coils	12
2.2.4	List of options and accessories	13
2.3	Phantoms	15
2.4	Quench tube	16
2.4.1	Quench tube inspection	16
2.4.2	Internal quench tube inspection	16
2.4.3	External quench tube inspection.	17
2.5	Magnet	20
2.5.1	Quench valve.	20
2.5.2	Ice formation check.	21
2.5.3	Leak check.	21
2.6	Patient fan	29
2.7	User documentation	31
2.8	User icons, button labels.	32
2.9	Identification of the controlled access area	33
2.9.1	Identification of the controlled access area (0.5 mT area)	33
2.9.2	Fringe field Area	34
2.9.3	Fringe field Skyra.	37
2.10	RF room door.	41
2.10.1	Door lock	41
2.11	Warning signs	42
2.11.1	Laser Warning Labels.	42
2.11.2	Option Combi Dockable Table: Warning labels	43
2.11.3	Tim Dockable Table Warning Labels	48
2.11.4	Hearing protection sign.	49
2.11.5	Warning Signs of the Pediatric Coil (Option).	50
2.12	End of General Tests	52
3	Electrical Tests	53
3.1	Measurement of the ground wire	53
3.1.1	Measurement of the system ground wire (every 2 years)	53
3.1.2	Measurement of the fixed Patient Table Ground Wire (every 2 years).	55

3.1.3	Measurement of the Dockable Patient Table Ground Wire (annually)	56
3.1.4	Measurement of the MRSC (MRWP) ground wire (annually)	57
3.2	End of Electrical Tests	58
4	Function Tests	59
4.1	Emergency power off switches	59
4.2	Gradient Supervision	60
4.3	Patient table	67
4.3.1	STOP function	67
4.3.2	Emergency release	69
4.3.3	Bruising hazard on the service end	70
4.3.4	Distance between the patient table and the cover	72
4.3.5	Table movement	73
4.3.6	Squeeze bulb	77
4.3.7	Option dockable table: pressure activated switches	78
4.3.8	Option dockable table: emergency undock function	79
4.3.9	Option dockable table: chassis and wheels	79
4.3.10	Option dockable table: lowering level of the undocked table . . .	80
4.4	ERDU	82
4.4.1	Prerequisites	82
4.4.2	Testing the Magnet Stop button (ERDU)	82
4.4.3	Final steps	86
4.4.4	ERDU Test status	86
4.5	Combi Dockable Table	88
4.5.1	Option Combi Dockable Table: STOP Function	88
4.5.2	Option Combi Dockable Table: Emergency Release	89
4.5.3	Option Combi Dockable Table: Distance between the patient table and the cover	90
4.5.4	Option Combi Dockable Table: Table movement	91
4.5.5	Option Combi Dockable Table: Squeeze bulb	95
4.5.6	Option Combi Dockable Table: Pressure activated switches	96
4.5.7	Option Combi Dockable Table: Emergency Undock Function . . .	97
4.5.8	Option Combi Dockable Table: Chassis and Wheels	98
4.5.9	Option Combi Dockable Table: Lowering level of the undocked table	99
4.5.10	Option Combi Dockable Table: Checking the interface box	100
4.5.11	Option Combi Dockable Table: Floor Dock	102
4.6	Quality Assurance (QA)	104
4.7	End of Function Tests	105
5	Changes to Previous Version	106
6	List of Hazard IDs	107

1.1 Safety Information



WARNING

Please follow the general safety guidelines as well as the MR-specific safety information!

Failure to comply with the safety notes may result in death or serious injury.

- » Read the safety notes contained in the general safety notes as well as the MR-specific safety information in detail prior to starting work. Adhere to the information provided at all times.

1.2 General remarks

Based on the risk analysis and the clinical evaluation, the manufacturer is obligated to determine the type and frequency of repair activities as well as the inspection measures required for products to ensure continuously safe and proper operation.

The safety-related tests

- are recurring tests. The scope and schedule of these tests are defined by the manufacturer.
- should determine whether the affected product is in good operating condition, ensuring the safety of patients, users, and other individuals.
- do not replace maintenance activities.

Country-specific laws and regulations have to be taken into consideration, e.g. in Germany the "Medizinprodukte-Betreiberverordnung (MPBetreibV)" (Medical Device Operator Regulations), IEC 62353 / DIN EN 62353, etc.

These safety-related tests correspond to the technical safety checks according to "MPBetreibV" (Medical Device Operator Regulations) "Sicherheitstechnische Kontrolle" including IEC 62353 / DIN EN 62353.

The safety-related checks have to be performed **during installation** and at the **intervals** defined in the time schedule.

1.3 Documentation

A protocol of the safety-related tests has to be generated. All fields have to be completed. If a field does not apply to the system or if there is no information to be entered, enter 'n.a.' in the field.

Each test performed successfully has to be confirmed in the test protocol by placing a check mark next to "O.K." Tests that cannot be performed (e.g. options) or which are not performed (e.g. interval is more than one year) should be marked with "n.a." If an error occurs which cannot be eliminated, mark the respective test with "not O.K."

The protocol of the installation "First measurement" serves as a reference document for subsequent repeat measurements. For this reason, it has to be archived by the operator in the System Owner Manual.

The MR quality department needs the protocol for legal reasons. Send a copy of the original to:

Tab. 1 Address MR Quality Department

Siemens
Healthcare
MR Quality Department
P.O. BOX 3260
91050 Erlangen
Germany

mail to: immrqpinstallations.healthcare@siemens.com

The protocols of repeat measurements should be filed in the System Owner Manual as well until subsequent safety-related tests have been performed.

We recommend that the country organizations also compile such documentation to ensure optimal operation of products e.g. in the event damages occur.

1.4 Acronyms:

Abbr.	Explanation
SI	Safety Inspection
SIE	Safety Inspection Electrical Safety
SIM	Safety Inspection Mechanical Safety
PM	Preventive Maintenance
PMP	Preventive Maintenance Preventive Parts Exchange, External Inspection, etc.
PMA	Preventive Maintenance Adjustments
PMF	Preventive Maintenance Function, Operating Value Check
Q	Quality Check
QIQ	Quality Check Image
QSQ	Quality Check System
SW	Software Maintenance

1.5 Time Frame for the Safety-Related Tests

Tab. 2 General tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
System	12	20	
Options and accessories	12	20	
Phantoms	12	10	
Quench tube	12	40	Quench tube outlet sign if missing, sign is part of the sign set 1-7 Adhesive tape for quench tube if missing, p.n. 3866493
Magnet	12	50	Helium leak detector
Patient fan	12	5	Air filter p.n. 10568825
User documentation	12	5	
Icon and button labels	12	5	
Controlled access area	12	10	Sign if missing, sign is part of the sign set 1-7
RF room door	12	5	
Warning signs	12	10	
Warning signs of the Pediatric Coil (Option)	12	5	Sign if missing; refer to SPC
Annual average time including all options: 185 min (3.1 h)			

Tab. 3 Electrical tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the system ground wire	24	30	Ground test meter
Measurement of the fixed patient table ground wire	24	5	Ground test meter
Measurement of the dockable patient table ground wire	12	5	Ground test meter

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Measurement of the MRSC ground wire	12	10	Ground test meter
Annual average time including all options: 30 min (0.5 h)			

Tab. 4 Function tests

Component / activity	Interval (month)	Time required (min)	Required tools and parts
EPO (if installed)	12	15	
Gradient supervision	12	40	Smoke wick p.n. 10684293 or Test spray p.n. 10684091
Patient table	12	60	
ERDU	12	15	
QA	12	70 - 110 ¹	QA Phantom set
Annual average time including all options ² : 220 min (3.6h)			

1. Depending on installed options (calculation with 90 min)

2. Depending on installed options (Without option Combi Dockable Table)

Tab. 5 Option: Combi Dockable Table

Component / activity	Interval (month)	Time required (min)	Required tools and parts
Combi Dockable Table	12	100	Label if missing; label set p.n. 10850296

2.1 System

Required time: 20 min

Required tools: n.a.

SI Visual inspection of system

- Perform the general visual inspection of the system.
 - » The system may not contain damages that impair safety.
 - » There must be no undefined parts inside the magnet room. They could be magnetic. Call the user to clarify before any movement of those parts.
- Correct visible defects with respect to safety prior performing further service works.

SI Inspecting visible cables and cable guides

- Check visible cables and cable guides for damage.

2.2 Options and accessories

Required time: 20 min

Required tools: n.a.

- Perform a visual inspection of the options and accessories.
- In conjunction with the operator/user, determine whether additional options/accessories (e.g., a trolley from an old system) are in use.
 - » Options and accessories must not have sustained any safety-relevant damage.
- Correct visible defects with respect to safety prior to performing further service work.

In order to document which options have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

2.2.1 Tim Dockable Table (optional)

- Perform a visual inspection of the dockable patient table. Thoroughly inspect the frame, the wheels, interlocking mechanism for damage, and take corrective action, if necessary.

In order to document which Tim Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

2.2.2 Combi Dockable Table (optional)

- Perform a visual inspection of the Combi Dockable Table. Thoroughly inspect the frame, the handles, the wheels, and interlocking mechanism for damage. Take corrective action if necessary.

In order to document which Combi Dockable Table has been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

2.2.3 Surface coils

- Check housing, cables, and connectors.
- Cables and connectors must not be damaged.
- Remove dust or metal shavings from inside and outside the connectors.

In order to document which coils have been inspected, enter the material and the serial number of the inspected options in the Safety-Related Tests Report (option and accessory list).

2.2.4 List of options and accessories



This section is to provide a visual and physical inspection of Options and Accessories to ensure safe operation. The check should be performed on Options and Accessories, such as Coils, Workstations, trolleys, etc., that were provided by Siemens with Siemens Material and Serial numbers. This check should only be performed on items located at the system undergoing maintenance.

- SI Head / Neck coil
- SI Spine coil
- SI Body 18 coil
- SI PA coil
- SI Shoulder Coil
- SI Wrist coil
- SI Foot Ankle coil
- SI Loop coil
- SI Flex coil small
- SI Flex coil large
- SI Knee coil
- SI Breast coil
- SI Tim-Adapter
- SI Extremity coil

2.2.4.1 Option MRSC

- SI Option MRSC

2.2.4.2 Dockable table option

- SI Dockable table option

2.2.4.3 Option 1



If an option (for example an another coil) is not listed in the protocol, enter this coil in the field Option 1 - 5.

- SI Option 1

2.2.4.4 Option 2

SI Option 2

2.2.4.5 Option 3

SI Option 3

2.2.4.6 Option 4

SI Option 4

2.2.4.7 Option 5

SI Option 5

2.3 Phantoms

Required time: 10 min

Required tools: n.a.

SI Checking the phantoms

Please read the “MR-Specific Safety Information” prior to inspecting the phantoms.

1. Inspect every phantom used on-site for cracks, leaks, or extensive wear and tear.
 - » Replace the phantom if there is any evidence of damage that may affect normal operation.



Defective phantoms must be disposed of. Refer to the “Disposal Instructions”.

2. Inspect every phantom used on-site for excessive air bubbles. Air bubbles can influence your Tune-up results and quality measurements.
3. Ensure that fluid is at an adequate level so that measurement performance is not adversely affected.

Water mixture phantoms with filler cap:

- Refill the phantom, if necessary.
- Phantoms with distilled water mixture - fill with distilled water only.

Water mixture phantoms without filler cap (welded):

- Refilling is not possible. Water mixture phantoms without filler cap must be replaced

Oil phantoms without filler cap (welded):

- Refilling is not possible. Oil phantoms without filler cap must be replaced

Oil phantoms with filler cap:

- Refilling is not possible. Oil phantoms with filler cap must be replaced.

Do not open the screw cap. It will lead to leakage.

For example spherical phantom blue D240 p/n 7577666 or p/n 10496685 : The diameter of the air bubble should not exceed 12 cm.

2.4 Quench tube

2.4.1 Quench tube inspection

Required time: 40 min

Required tools: If missing from the quench tube the following spare parts are available from the SPC and must be fitted:-

- Adhesive tape for Quench tube p.n. 3866493
- Quench outlet warning sign.

SI Visual inspection of the quench tube

- Visually inspect the condition of the quench tube.

2.4.2 Internal quench tube inspection

Check the following (visible inspection from the ground only, it is not necessary to remove ceiling tiles etc):

- Check that no unauthorized changes to the design of the quench tube **have been made**, both inside or outside the RF room.
- Check the **quench tube** and **insulation** for **damage**.
- Check that the **connection screws** to the magnet quench valve are all fitted and tight.
- Check the warning **label or tape** is attached to the **tube insulation**.
 - **On long tube sections** the warning labels or tape should be at regular distances of **between 2 - 3 meters**.

Fig. 1: Warning label example- Vent tube

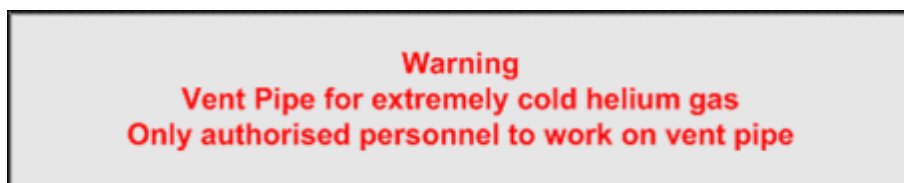


Fig. 2: Warning tape- Quench tube



Example of new warning tape supplied with magnets. This will provide a consistent warning message.

2.4.3 External quench tube inspection



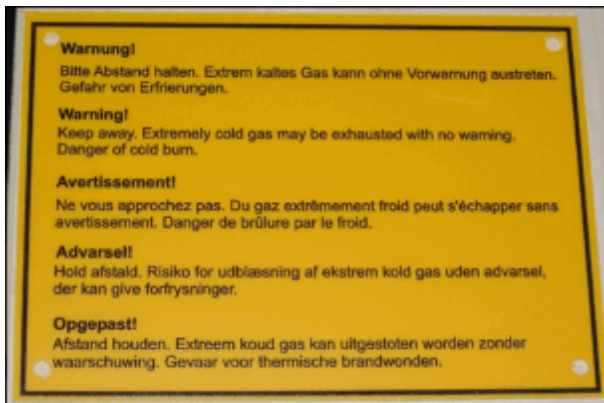
Warning Signs

The outlet warning sign must be in the National language.

Fig. 3: Warning sign example - Quench tube outlet



Fig. 4: Warning sign example - Quench tube outlet



Example of a new warning sign that provides a consistent message in several languages.

- Check the **warning sign** is fixed and clearly visible next to the **outlet** of the quench tube.
- Check the **outlet** of the quench tube for **obstructions** (e.g. bird's nest).
- Check for a **protective mesh** on the **quench tube outlet** and its integrity.
- Check a **protective rain cover** is fitted to the **vertical outlets**.
 - The cover must not be **damaged** or restrict the gas flow from the quench tube.

The **outlet** tube cannot be situated where, in case of a quench:-

- Helium gas can be drawn into an **air inlet**.
- Helium gas can enter **open windows**.

To avoid risk of **injury** from cold burns and asphyxiation, access to the quench tube must be restricted by:-

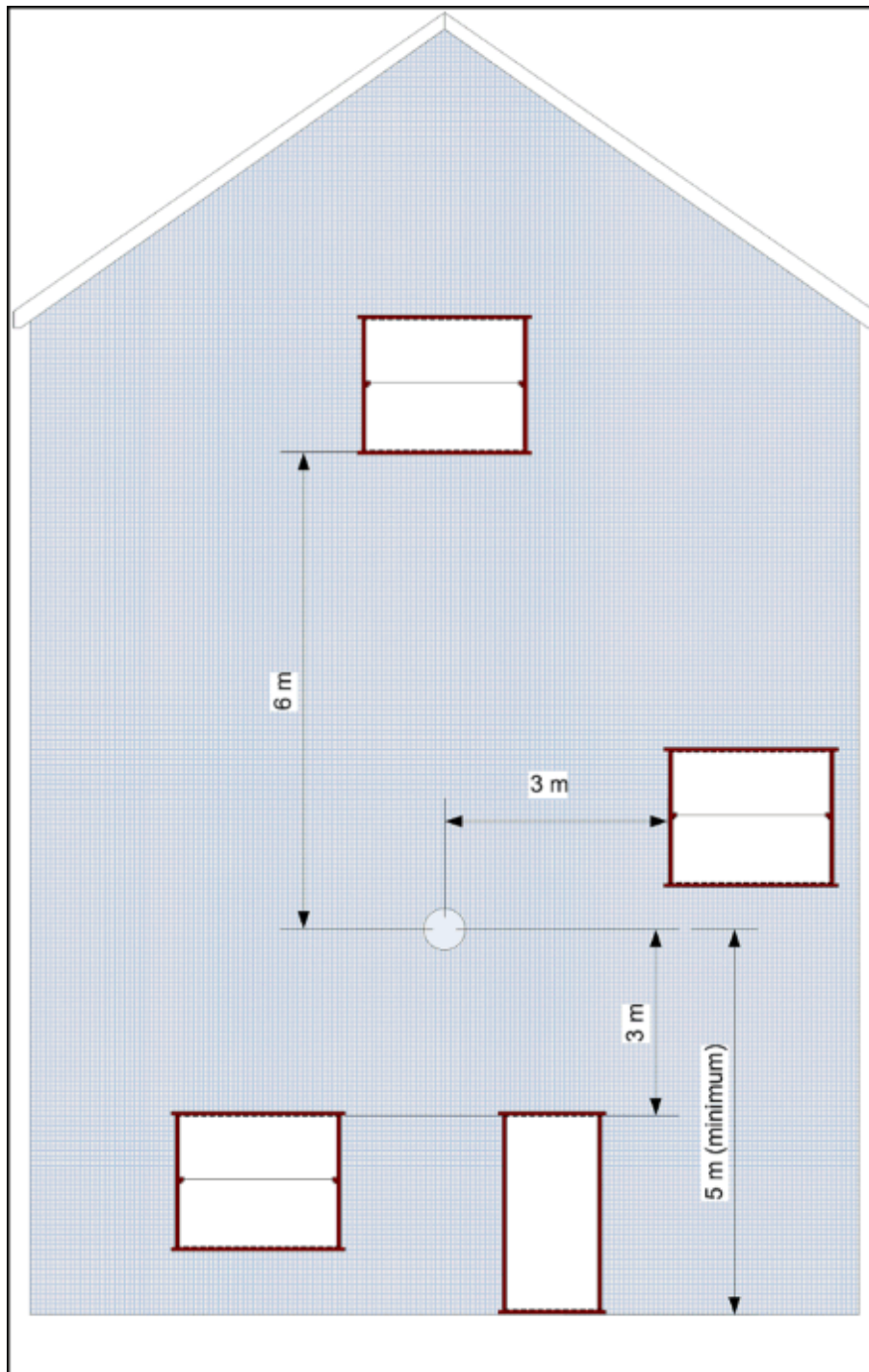
- 3 m on each side and below.(→ Fig. 5 Page 19)
- 6 m vertically above the exit.



Helium is lighter than air, even when very cold. The cold gas must not be allowed to blow directly onto a window.

- Check that the restricted area has not changed.
- Check that **venting gas** does not represent a hazard to people. For example:-
 - Subsequent building of windows
 - Air conditioning system intake or exhaust pipes
 - Erection of new or temporary buildings
 - Temporarily storing containers.
- If windows are present in the restricted access area, they must be **sealed** and **permanently closed**. Means of opening the windows must be **removed**.
- Inform the operator to initiate corrective action **immediately** in case you find obvious **defects** or problems which could restrict the flow of gas from the quench **tube**.
- Record your findings in the **Protocol**.

Fig. 5: Quench tube - example of venting to the outside



2.5 Magnet

Required time: 50 min.

Required tools: Electronic helium leak tester (material number: 10614850).

Prerequisites: Turret cover is removed (please refer to instruction > Replacement of Parts > System Covers)

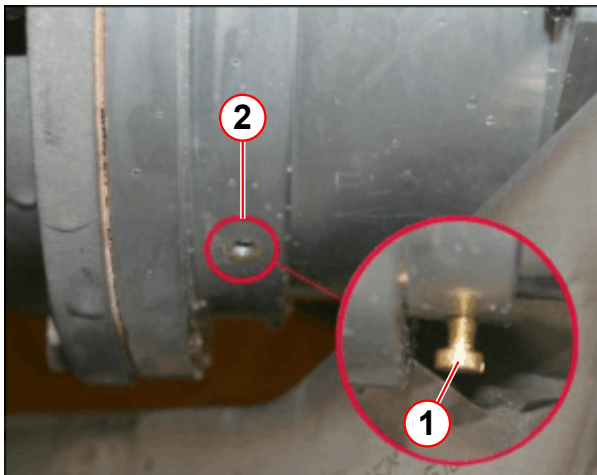
2.5.1 Quench valve

SI Check for water in the quench valve



Always use the magnet service platform when working on the service turret.

Fig. 6: Quench valve drain screw



- (1) M5 Metal screw
- (2) Drain screw hole location

The quench valve must be inspected for water.

- Check that the drain screw is **re-moved**.
- Check the drain hole is clear to allow water to drip out.

Fig. 7: Top cover removal



- With cables tied to the quench pipe, access to the drain hole can be difficult:
 - Remove the turret covers (left and right turret cover) to get access.

Fig. 8: Locating the drain hole



- It should be possible to feel the indent of the hole at the bottom of the quench valve.
- If necessary use an inspection mirror to view the drain hole.
- Use a 'cannula' or similar small component to check that the drain hole is clear of debris.



If water is present, determine the cause (e.g. damaged or missing protective rain cover, missing, quench tube insulation - leading to condensation). Advise the customer to initiate corrective action immediately.



For Static systems, the drain hole is permanently open - it allows water to drip freely from the quench valve.

On completion of the quench valve checks:

- Replace the system covers.

2.5.2 Ice formation check



Always use the magnet service platform when working on the service turret.

SI Check for ice formation at the magnet service turret and venting system

- The service turret and venting system should be free of ice formation (except after ramping up or filling).
 - » Empty out the water from the condensation reservoir located on the floor at the electronics side of the magnet.

2.5.3 Leak check

SI Magnet system leak check

The magnet is fitted with a recondensing cold head and pressurization heater which maintains the pressure in the helium vessel at a constant **15.5 psi (A)**, providing there are **no leaks** in the venting system.

Gross leaks show up as a **low magnet pressure**. Small leaks may not show any change in pressure.

All leaks are measured as a steady **loss** in the liquid helium level.



Insufficient cooling power of the cold head could also result in helium loss.

If the heater average power = 0W and magnet pressure ~ 16psia (Aera) or 17psia (Skyra), then check the cold head performance.

- The helium level and magnet pressure must be monitored regularly either by using **Service Software** on the **MRC ((MRAWP) Host)**.



If the liquid helium level and the magnet pressure have remained constant with no helium loss, no further leak check activities are necessary.



If the magnet has low pressure, or there is evidence of slow helium loss, then a full turret and venting leak check must be carried out.

Leak checking must be performed with an electronic leak detector capable of detecting leaks, at least as small as **1 x 10⁻⁴ mbar l/s**.

Fig. 9: Edwards Leak detector



- Use the Edwards hand-held leak detector (part no **10614850**).

This is the **recommended equipment** for leaks **greater than 1 x 10⁻⁴ mbar l/s**.

NOTE: The hand-held device is calibrated in **ml/sec**:-

1 x 10⁻⁴ mbar l/s = 1 x 10⁻⁴ ml/sec at normal pressure and temperature.



The functionality of the Edwards leak detector can be compromised by magnetic fields.

Leak check all seals that have been subjected to prolonged low temperature and the subsequent build-up of frost. For example, seals which have been cooled to low temperature by helium transfers, helium vessel de-pressurization, or magnet ramping.

Before leak checking the following magnet components must be at **room temperature**.

- Turret.
- Venting valve assembly.

- Quench valve.

All **high-pressure** joints and seals must be checked when there is **low system pressure** and **liquid helium loss**.

- Cold head. (→ Cold head / Page 24)
- Turret top plate, siphon port and venting. (→ Turret and venting / Page 27)
- Venting valve assembly. (→ Venting valve assembly / Page 28)
- Pressure gauge assembly. (→ Pressure gauge assembly / Page 28)
- Connecting pipe couplings.



If no external leak is found but the system pressure is low, then it is possible that one of the internal valves is leaking. This can be caused by a cracked or broken burst disk, or a broken O-ring.

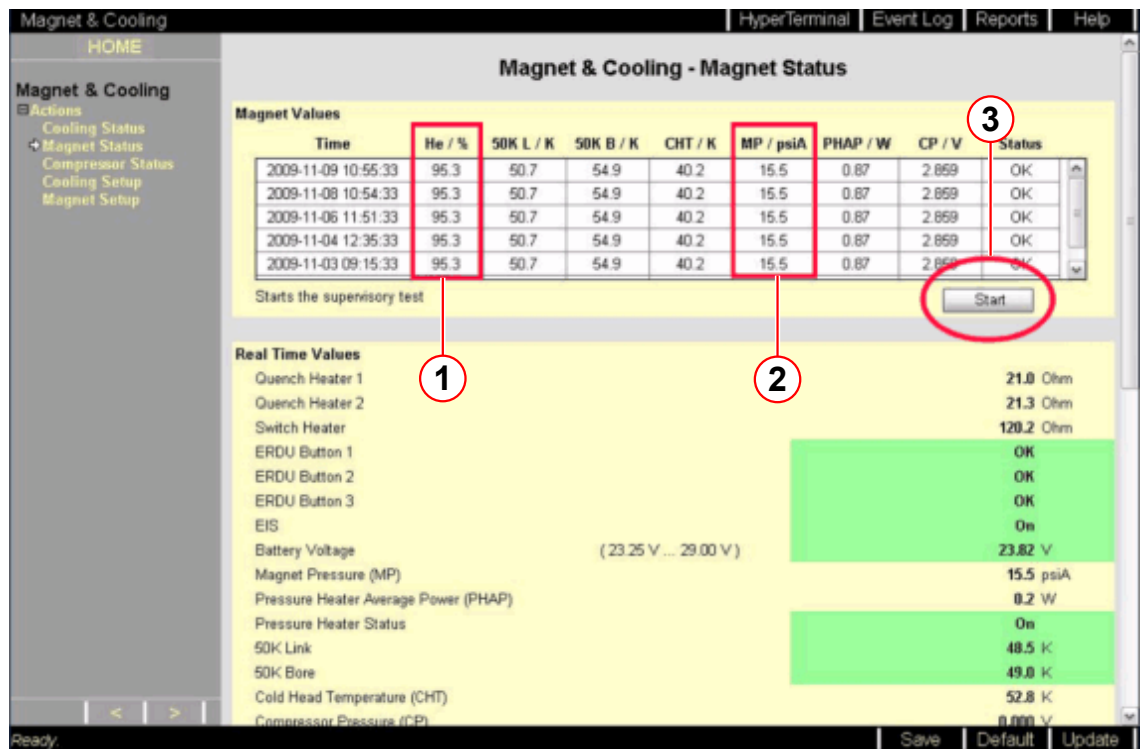


The magnet must be RUN TO ZERO before checking inside the quench valve or auxiliary vent.

- Check the liquid helium level by using Service Software on MRC (MRAWP) Host.
 - Check the liquid helium level using the MRC.
 - Access the Service Software at the MRC.
 - Select the menu item Magnet & Cooling - Magnet Status.

- Press the Start button (→ 3/ Fig. 10 Page 24) to initiate the helium measurement.
(→ 1/ Fig. 10 Page 24)

Fig. 10: Checking liquid helium and magnet pressure levels



- (1) Liquid helium level
- (2) Magnet pressure
- (3) Start button



Data is only accurate from the last refresh of the page, or from when the page was opened last.

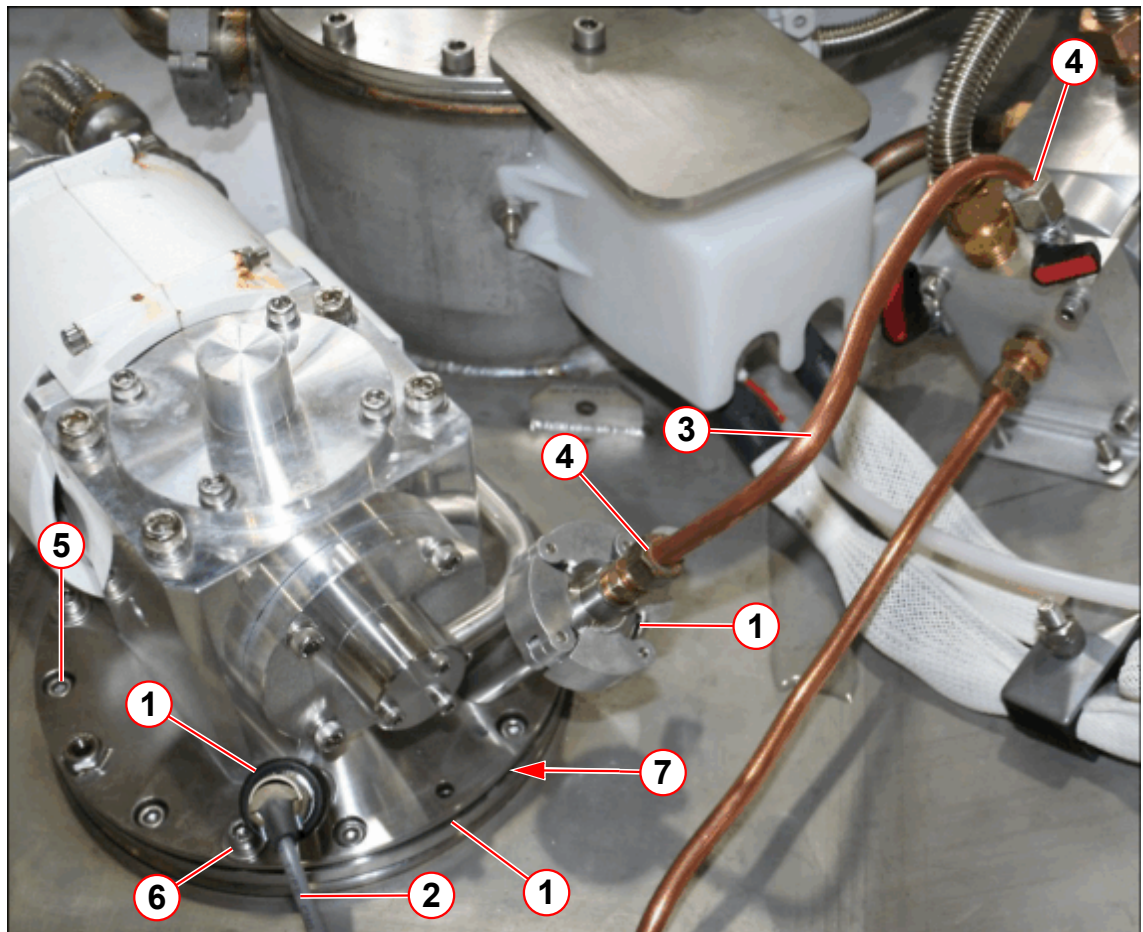
- The following checks must be carried out if the magnet has:
 - a **decreasing** liquid helium level. (→ 1/ Fig. 10 Page 24) or
 - a magnet pressure **below 15.5 psi (A)**. (→ 2/ Fig. 10 Page 24)

2.5.3.1 Cold head



The cold head vent line is to remain connected with the hand-valve CLOSED during normal system operation.

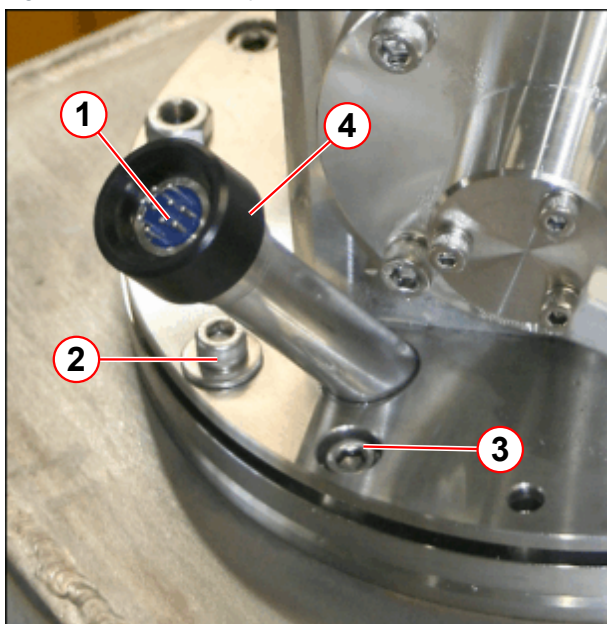
Fig. 11: Leak checking the cold head seals



- (1) Cold head seals leak check points
- (2) Temperature sensor cable
- (3) Cold head vent line
- (4) Cold head bypass line connection
- (5) 8 off M6 x 16 mm O-ring clamp ring screws, torque to = 8Nm
- (6) 4 off M6 x 25 mm cold head flange screws, torque to = 4Nm
- (7) O-ring seal

- Check the **vent line seals** (→ 1/Fig. 11 Page 25) at the **cold head** and **bypass line connection**. (→ 4/Fig. 11 Page 25)

Fig. 12: Cold head 10 pin seal



- (1) Leak check point
- (2) 4 off M6 x 25 mm cold head flange screws, torque to 4Nm
- (3) 8 off M6 x 16 mm skt hd screws O-ring clamp plate, torque to 8Nm
- (4) Black sealing cap

- **DO NOT** turn or try to remove the 'black' sealing cap
- Remove the temperature sensor cable (pos. 4), with a straight pull away from the black sealing cap.
- Leak check the connector and its seal.
- Immediately refit the cable if no leak is found.

The cold head is fitted with **temperature sensors**:

- The **feed-through** for the wires is sealed with an **O-ring**.
- Check the **feed-through** and **cold head seals**.
- **Check the torque** of the cold head **screws**.

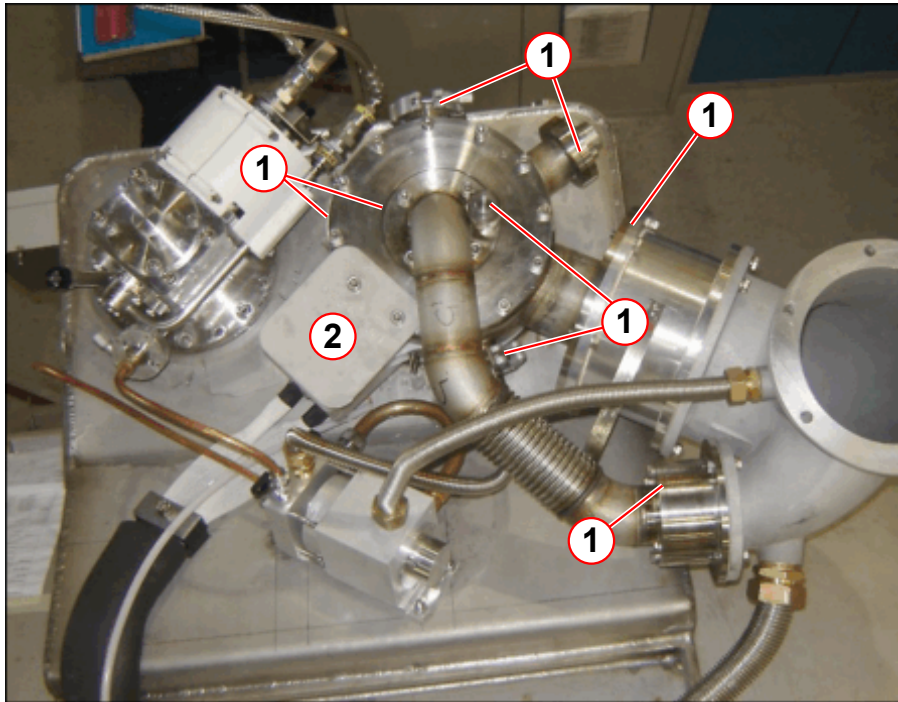


If a leak is found with the magnet at field, it is permissible to re-torque the 4 off cold head flange screws to a higher setting of 5 Nm.

If the leak persists, send a report to your local Uptime Service Center for repair by a specialist.

2.5.3.2 Turret and venting

Fig. 13: Turret and venting - high pressure seals



- (1) Turret top plate, syphon port and O-rings
- (2) Positive FCL connector (not visible)

- Check that all **clamps** on the **high-pressure side** of the venting are tight. **Do not** try to **over-tighten** them - it could cause a leak, or make an existing leak worse.



If no leak is found, but there is a slow, steady liquid helium loss, contact your local Up-time Service Center. Specialist repair is required.



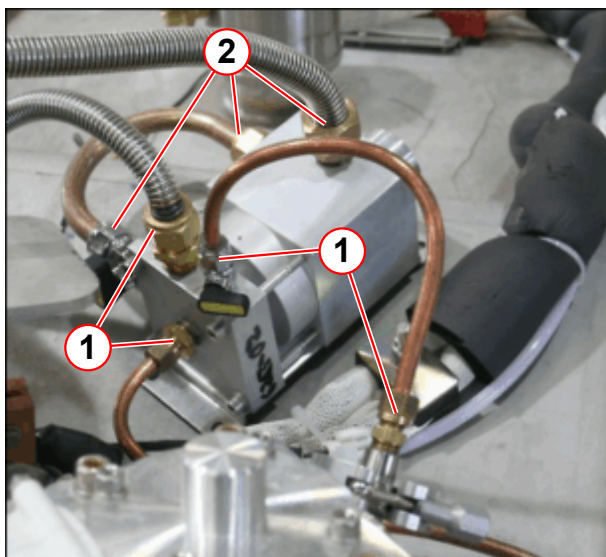
Do not use any form of vacuum grease on O-rings and seals.
It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.



Replace a high load seal only with another identical high load seal. High load seals must only be used **ONCE** for a guaranteed leak tight seal.

2.5.3.3 Venting valve assembly

Fig. 14: Vent valve assembly leak check points

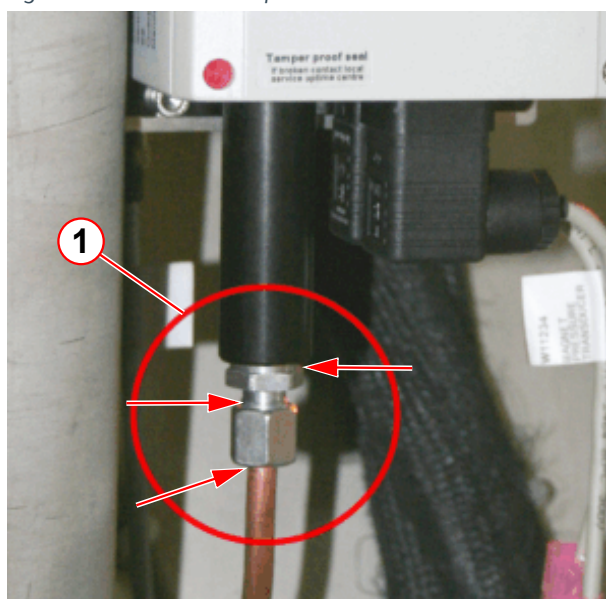


- (1) High-pressure joints
(2) Low-pressure joints

- Leak check the venting valve **high pressure** assembly.

2.5.3.4 Pressure gauge assembly

Fig. 15: Leak check the pressure line



- (1) Leak check points

- Leak check the **high pressure** points.

Contact your local **Uptime Service Center** if further leaks are suspected.

2.6 Patient fan

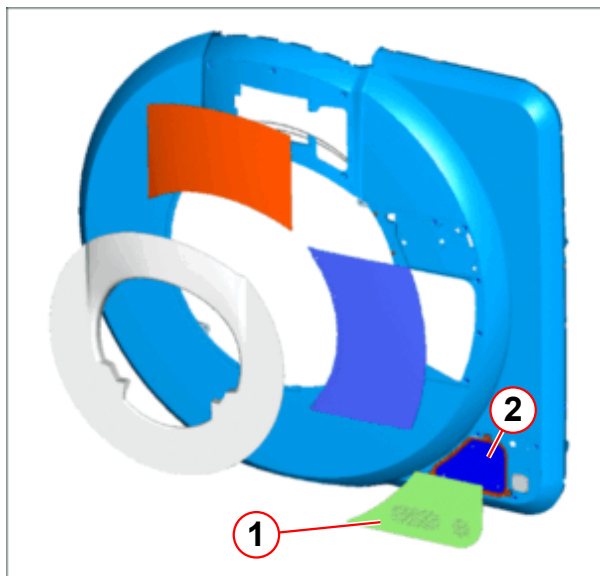
SI Checking the air filter of the patient fan

Required time: 5 min

Required tools and parts: Air filter (material number 10568825)

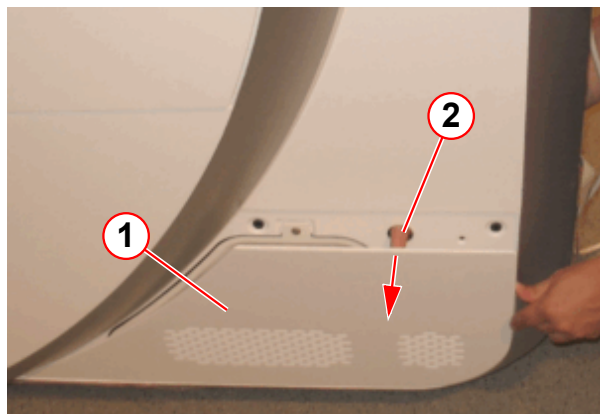
- Remove the service cover.

Fig. 16: Rear cover



- (1) Air filter cover
- (2) Air filter

Fig. 17: Cover - service end



- (1) Cover - patient air filter
- (2) Hole to press out the air filter cover

The air filter cover is fixed with snap locks. A hole (→ 2/ Fig. 17 Page 29) is located in the rear cover to press out the air filter cover.

- Press out the air filter cover and remove it.

Fig. 18: Filter of the patient air fan



- (1) Filter pad
- (2) Hole to press out the air filter cover

- Remove the filter
- Install the new air filter.

- Install the air filter cover in reverse order.

2.7 User documentation

Required time: 5 min

Required tools: n.a.

SI User documentation available and legible

- Ensure that the System Manual is available, legible and valid (e.g. software version).

2.8 User icons, button labels

Required time: 5 min

Required tools: n.a.

SI Checking user icons and button labels

- Inspect the button labeling and user icons for legibility/recognizability (for example “symbol keypad”, “**Magnet Stop button**”, “Patient Table Stop”, etc.).

2.9 Identification of the controlled access area

2.9.1 Identification of the controlled access area (0.5 mT area)

Required time: 10 min

Required tools: n.a.

Refer to the Spare Parts Catalogue for the order number for the set of warning signs.

SI Identification of the controlled access area (0.5 mT area)

Fig. 19: Warning sign "Strong Magnetic Field"



Fig. 20: Warning sign "Cardiac Pacemaker"



Prior to initial Startup, the project manager defines the 0.5 mT area.

- Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone. The signs have to be posted in a way that it is visible even when the door to the controlled area is opened.

SI Checking the warning signs (SIEMENS)

SI Checking the warning signs (customer signs cover all safety aspects as listed on the SIEMENS signs)

The fringe field extends three-dimensional beyond the magnet and can be reduced through shielding. The field lines represent the ideal flux density distribution in air, which may be distorted by steel reinforcement in buildings.

2.9.2 Fringe field Aera

Fig. 21: Fringe Field distribution Y-Z Direction

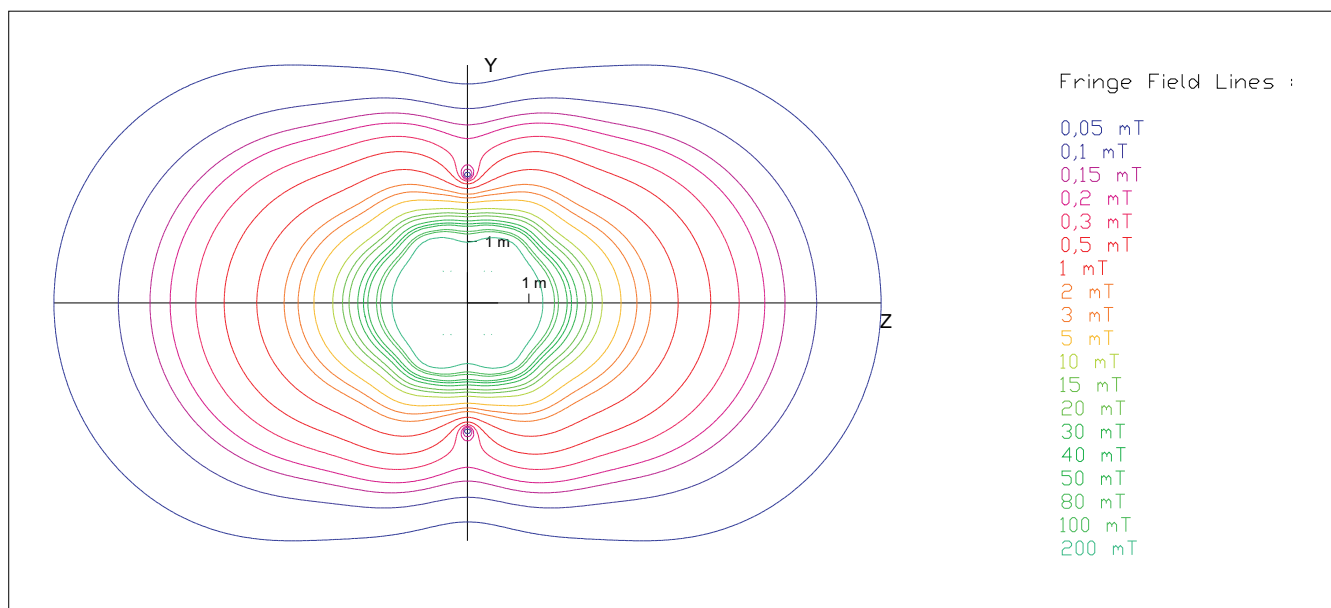


Fig. 22: Fringe Field distribution Y-X Direction

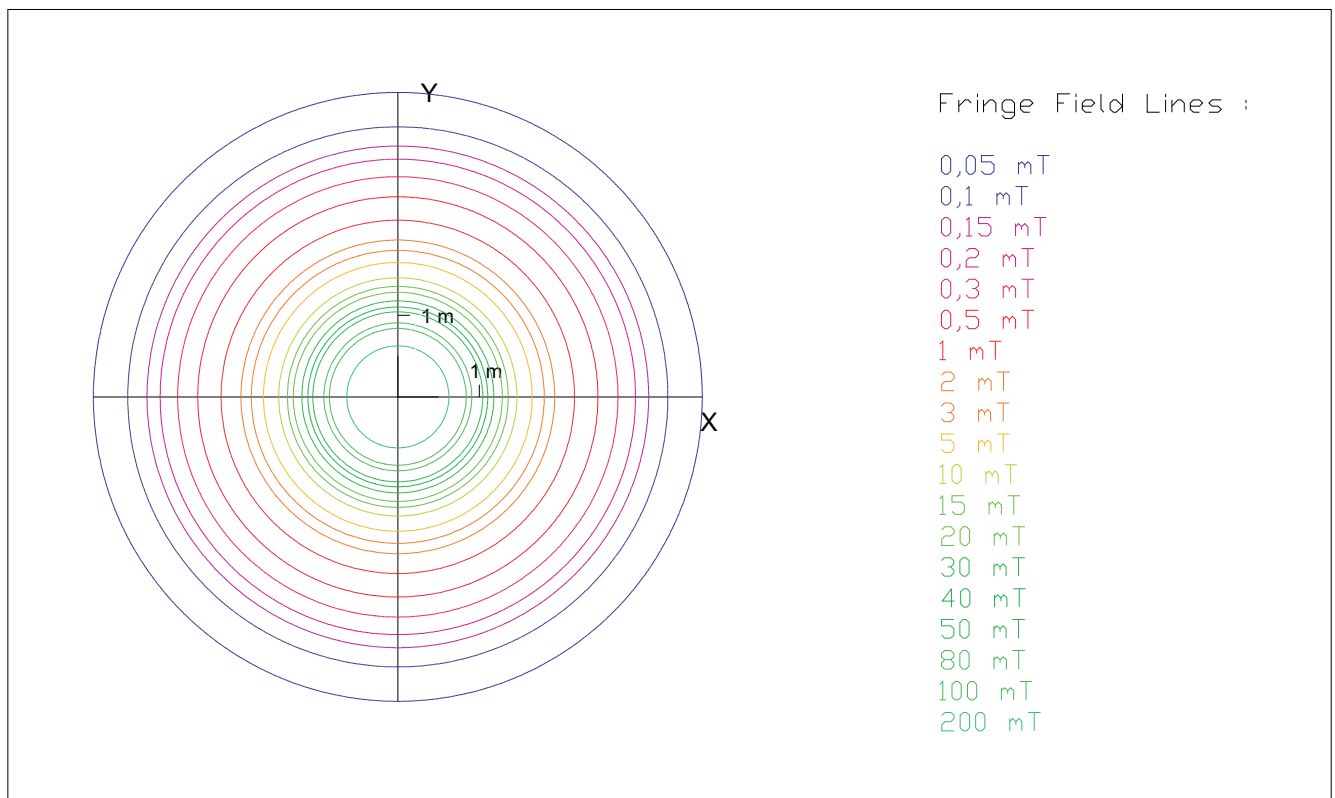
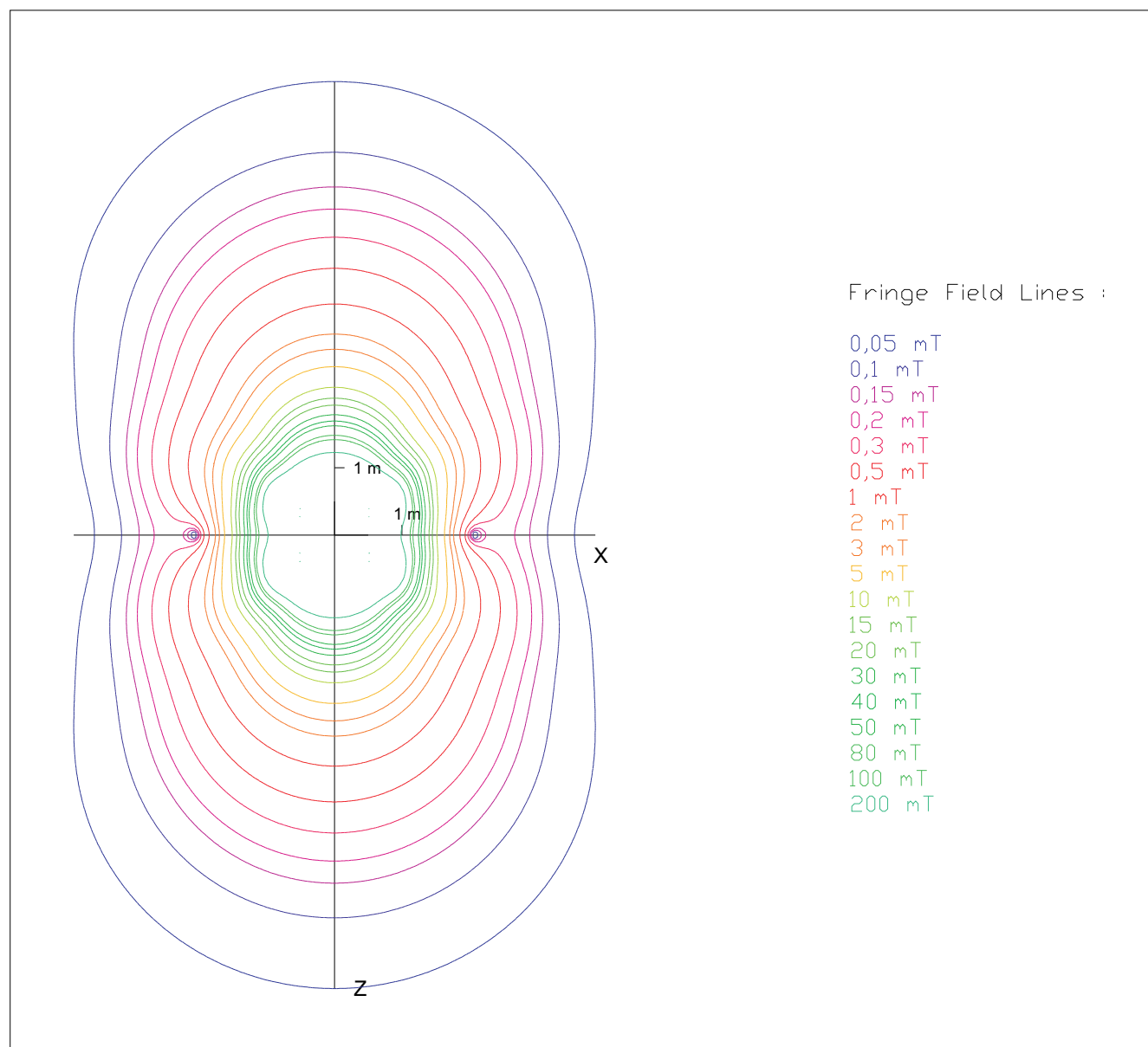


Fig. 23: Fringe field distribution Z-X Direction



Tab. 6 Fringe field distribution MAGNETOM Aera

Fringe field	Distance from the magnet isocenter in the direction of		
	the X-axis in [m]	the Y-axis in [m]	the Z-axis in [m]
40 mT	1.3	1.3	1.7
20 mT	1.4	1.4	2.0
10 mT	1.5	1.5	2.2
5 mT	1.7	1.7	2.5
3 mT	1.8	1.8	2.8
1 mT	2.2	2.2	3.4

0.5 mT	2.50	2.50	4.00
0.3 mT	2.7	2.7	4.4
0.15 mT	3.1	3.1	5.2
0.1 mT	3.2	3.3	5.7
0.05 mT	3.9	3.9	6.7

2.9.3 Fringe field Skyra

Fig. 24: Fringe Field Distribution Z-Y Direction

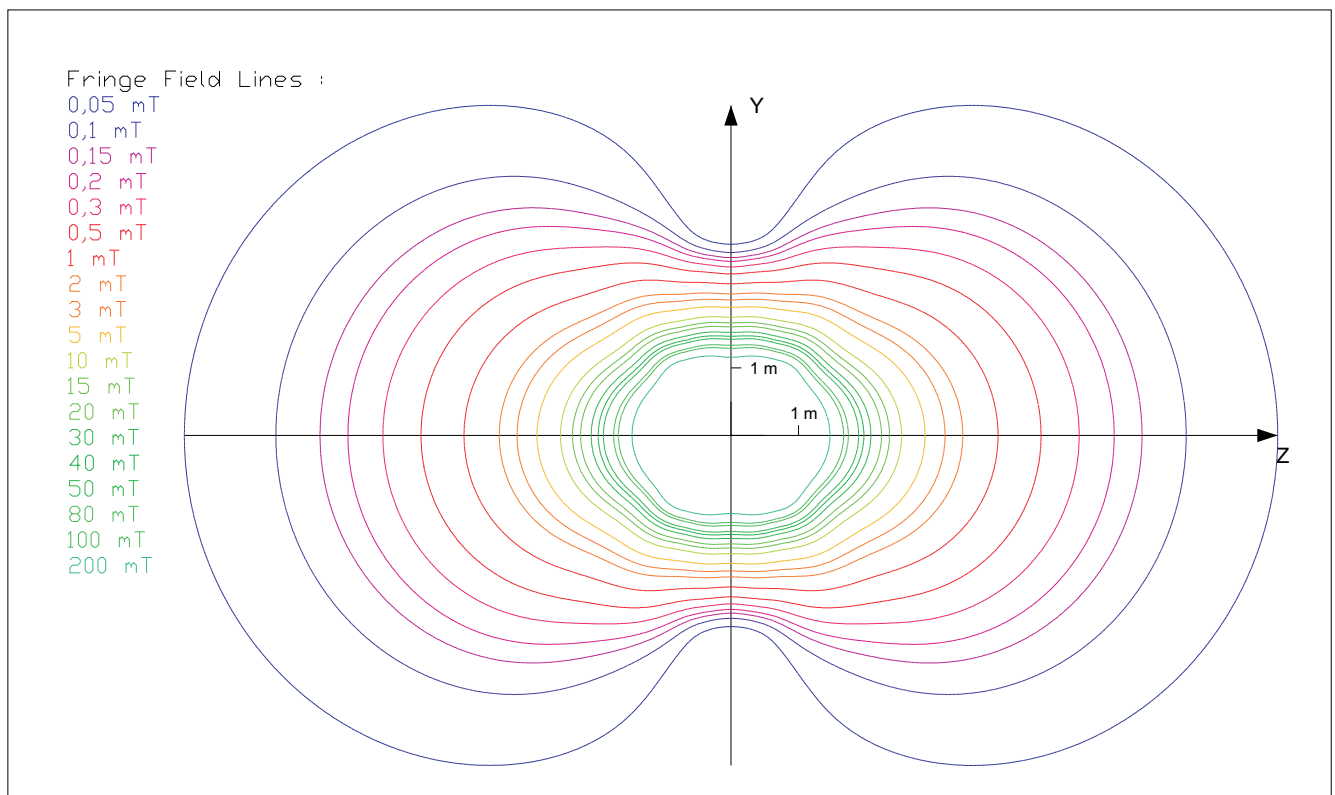


Fig. 25: Fringe Field Distribution Z-X Direction

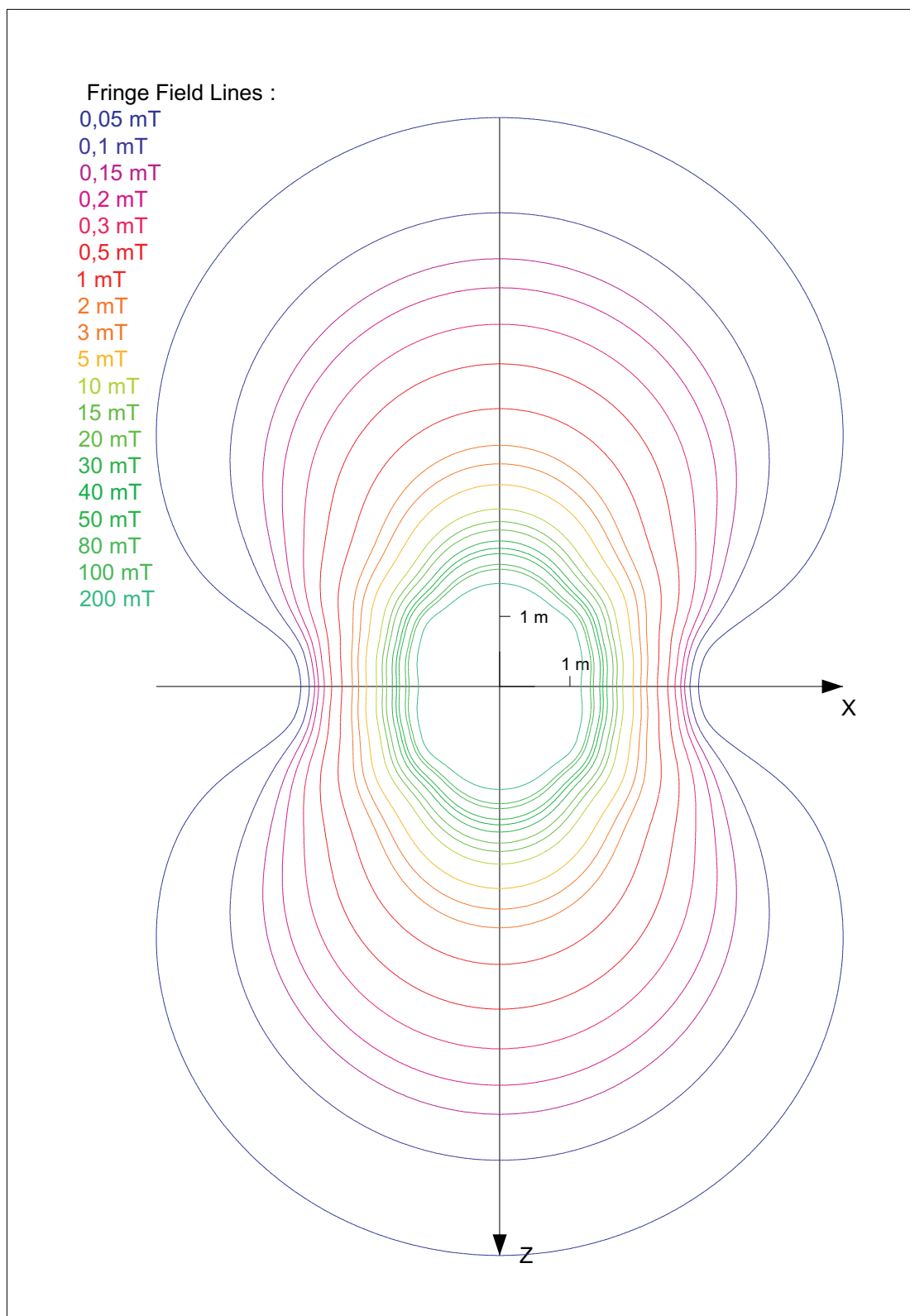
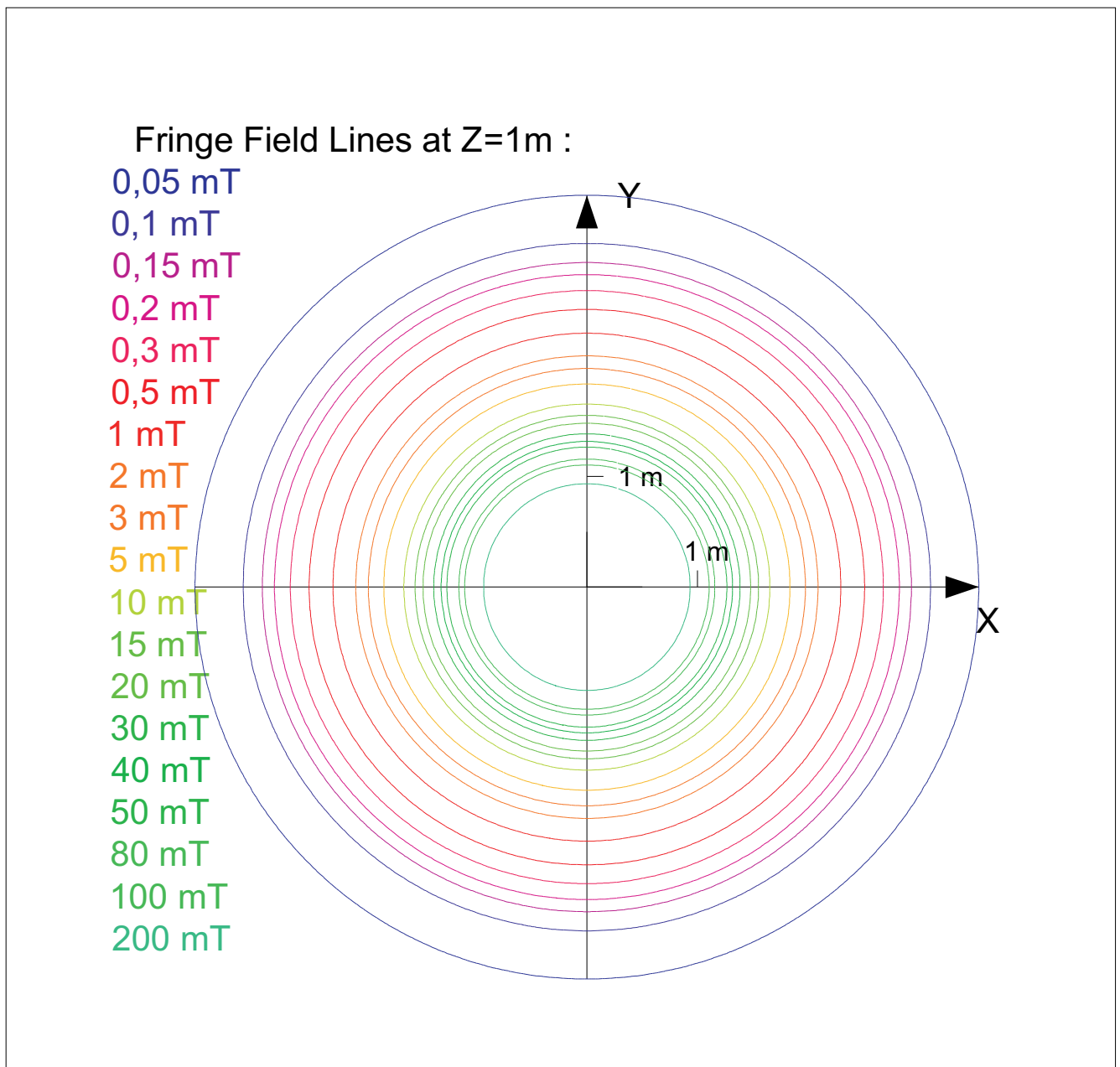


Fig. 26: Fringe Field Distribution X-Y Direction



Tab. 7 Fringe field distribution MAGNETOM Skyra

Fringe field	Distance from the magnet isocenter in the direction of		
	the X-axis in m	the Y-axis in m	the Z-axis in m
40 mT	1.5	1.5	2.0
20 mT	1.6	1.6	2.2
10 mT	1.8	1.8	2.5
5 mT	1.9	1.9	2.9
3 mT	2.1	2.1	3.2

1 mT	2.3	2.3	4.0
0.5 mT	2.60	2.60	4.60
0.3 mT	2.8	2.8	5.2
0.15 mT	3.4	3.4	6.1
0.1 mT	3.9	3.9	6.8
0.05 mT	4.9	4.9	8.2

2.10 RF room door

Required time: 5 min

Required tools: n.a.

2.10.1 Door lock

SI Checking the door lock of the RF room

1. The locking and unlocking mechanism has to function properly.
2. The door handle or knob has to be securely fastened.



In case of a defect (e.g. defective unlocking mechanism, door handle,), inform the customer to initiate corrective action immediately.

2.11 Warning signs

2.11.1 Laser Warning Labels

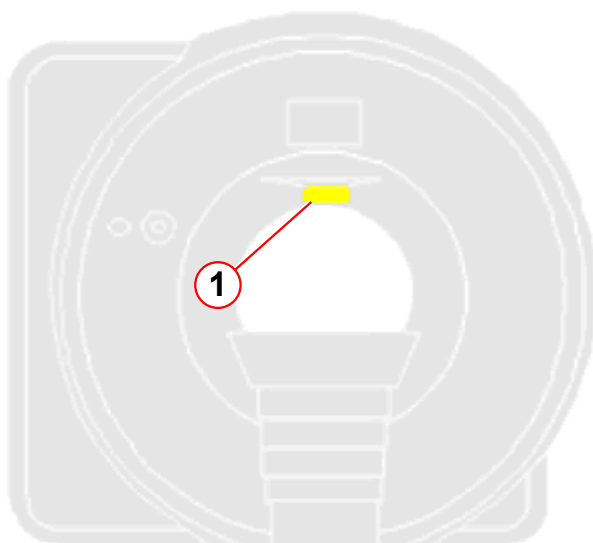
Required time: 2 min

Required tools or material: warning labels, if missing

The warning labels are part of the label set 098 material number: 10568937

SI Checking the laser warning labels

Fig. 27: Magnet patient end



(1) Laser label

- Ensure that the correct warning labels are affixed to the cover (→ 1/Fig. 27 Page 42).
If the "3rd control panel" (magnet service end) option is installed, ensure that the correct warning labels are affixed to the cover.
 - » The labels must be legible.
- Replace missing labels immediately.

2.11.1.1 Label Configuration for the US Certification Label, warning logo type and aperture label

Fig. 28: Laser label configuration USA



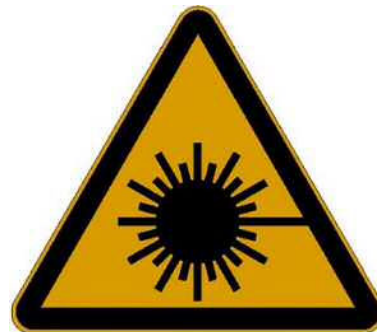
- (1) Aperture label, warning logo type
- (2) Certification label

2.11.1.2 Label Configuration for all other countries is either in the local language or an alternative one.

Fig. 29: Laser label 1 (IEC 60825 - 1:2007)



Fig. 30: Laser label



2.11.2 Option Combi Dockable Table: Warning labels

Required time: 5 min

Required tools or material: warning labels p.n. 10850296 , if missing

SI Option Combi Dockable Table: Checking the warning labels

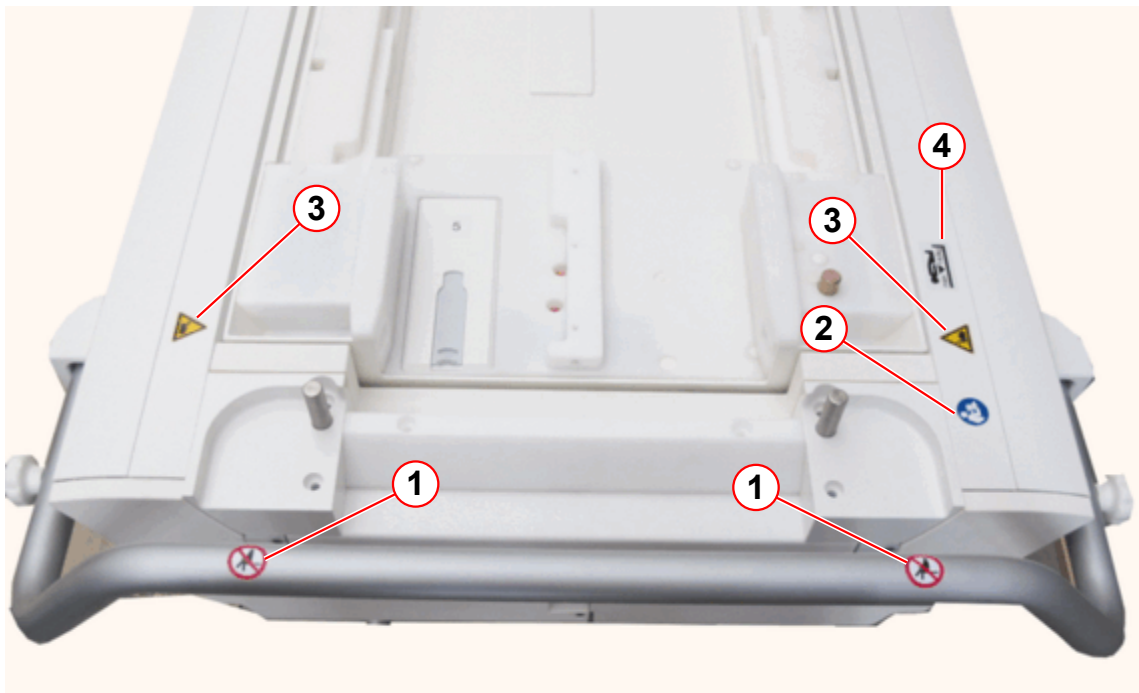
- Ensure that the warning labels are affixed to the Combi Dockable Table.
- Replace missing labels immediately.
 - » The labels must be legible.



Three types of the interface box are available depending on the different tables and transfer boards used for transfer to these tables. The label configuration at the foot end differs according to the interface boxes.

- Label configuration at the foot end: TRUMPF interface box

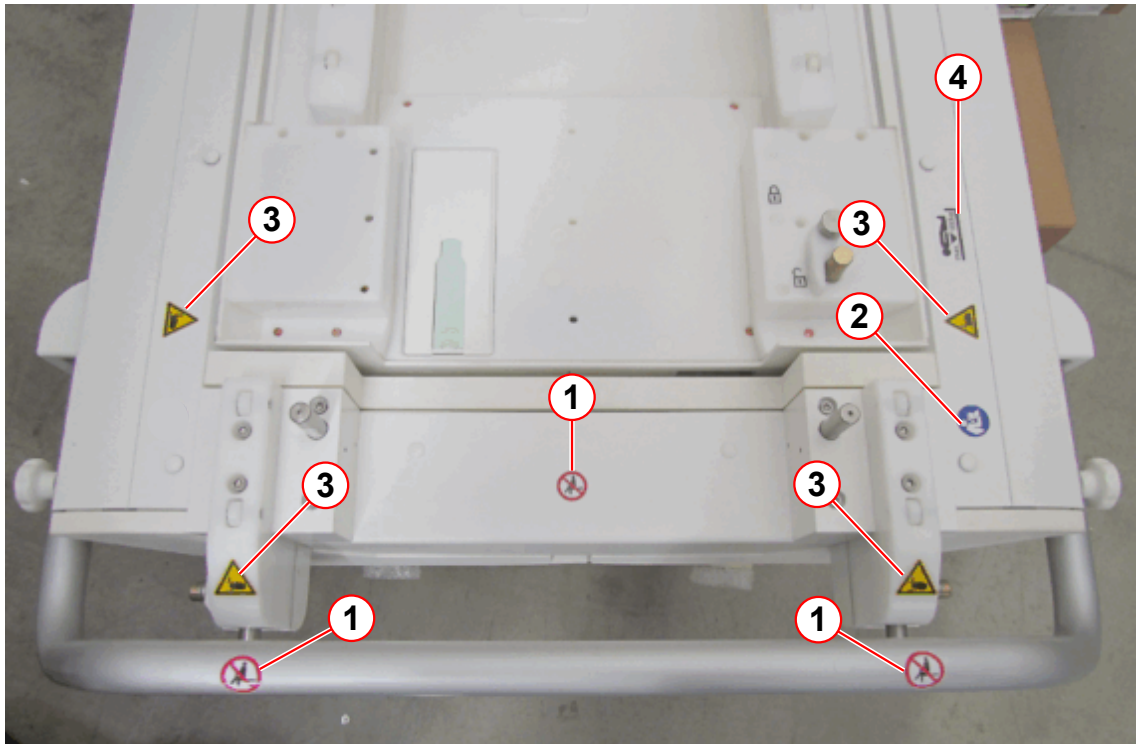
Fig. 31: Labels at the foot end: TRUMPF interface box



- (1) Label - Do not sit
- (2) Label - Observe manual
- (3) Label - Bruising hazard
- (4) Label - max. load 140 kg

- Label configuration at the foot end: MAQUET interface box

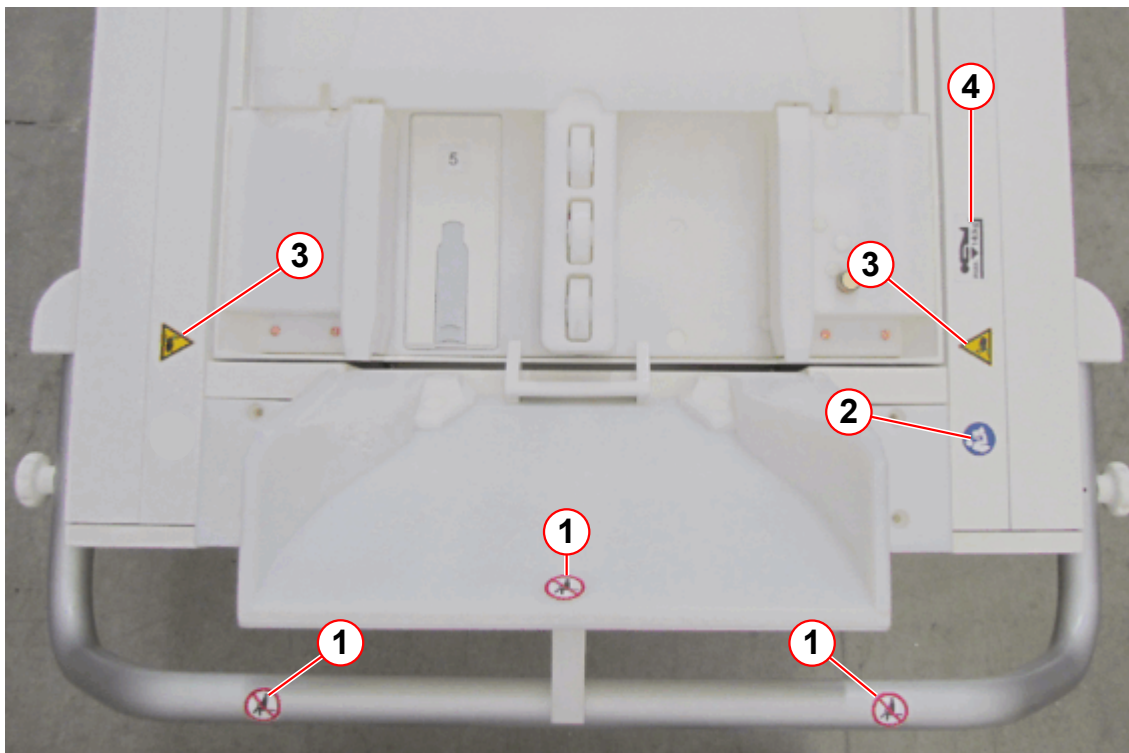
Fig. 32: Labels at the foot end: MAQUET interface box



- (1) Label - Do not sit
- (2) Label - Observe manual
- (3) Label - Bruising hazard
- (4) Label - max. load 180 kg

- Label configuration at the foot end: Artis interface box

Fig. 33: Labels at the foot end: Artis interface box



- (1) Label - Do not sit
- (2) Label - Observe manual
- (3) Label - Bruising hazard
- (4) Label - max. load 140 kg

- The label configuration at the head end is identical for all table types.

Fig. 34: Combi Dockable Table - head end



- (1) Label - Do not sit
(2) Label - 100N

Table warning labels

Fig. 35: Label - 100N



Fig. 36: Label - Do not sit



Fig. 37: Label - max. load 140 kg



Fig. 38: Label - max. load 180 kg

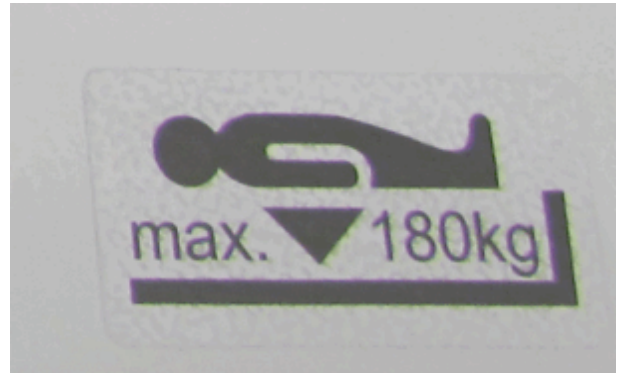


Fig. 39: Label - Observe manual



Fig. 40: Label - Bruising hazard



2.11.3 Tim Dockable Table Warning Labels

Required time: 2 min

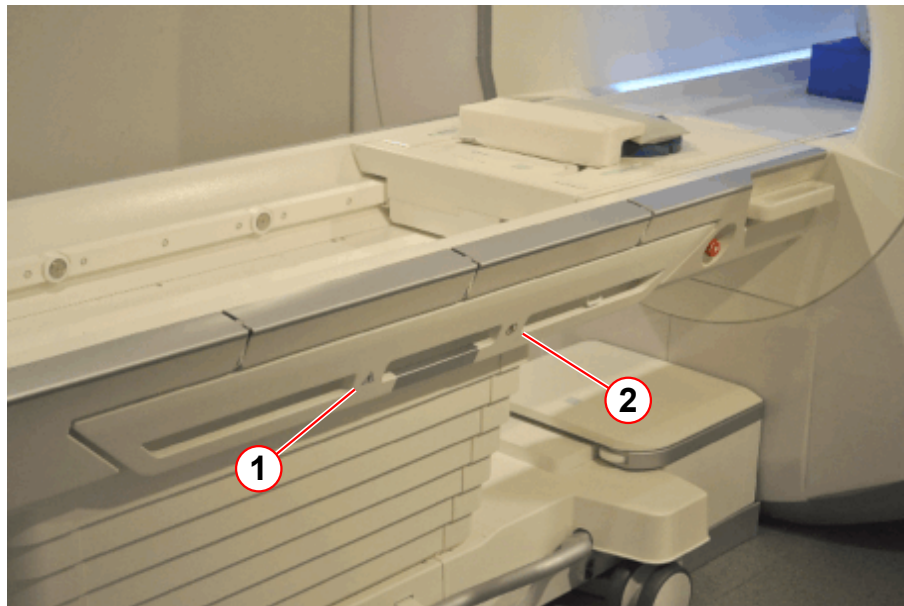
Required tools or material: warning labels, if missing

The warning labels are part of the label set 098 material number: 10568937

SI Checking the warning signs of the dockable table

- Ensure that the warning labels - (→ Fig. 43 Page 49) Do not sit and - (→ Fig. 42 Page 49) Maximum load 190N are affixed to the left and right side railing of the dockable table (→ Fig. 41 Page 49).

Fig. 41: Tim dockable table - side railing



- (1) Label - Maximum load 190N
- (2) Label - Do not sit

Fig. 42: Label - Maximum load 190N



Fig. 43: Label - Do not sit



- » The labels must be legible.
- Replace missing labels immediately.

2.11.4 Hearing protection sign

Required time: 3 min

Required tools or parts: if missing hearing protection sign. The hearing protection sign is part of the warning sign sets, please refer to SPC.

SI Checking the hearing protection sign

- Ensure that the hearing protection sign has been posted at all entrances to the examination room.

Fig. 44: Hearing protection sign



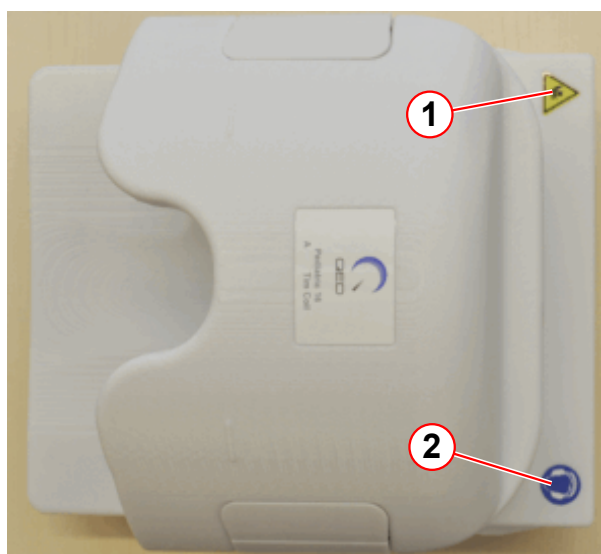
- Install missing signs immediately

2.11.5 Warning Signs of the Pediatric Coil (Option)

Required time: 5 min

Required parts and tools: Sign if missing.

Fig. 45: Pediatric coil



- (1) Sign for bruising hazards
- (2) Hearing protection sign

SI Checking the warning signs of the pediatric coil

- Ensure that the warning signs (→ Fig. 46 Page 51), (→ Fig. 47 Page 51) are affixed to the pediatric coil.

- Replace missing labels immediately.

Fig. 46: Sign for bruising hazards



Fig. 47: Hearing protection sign



2.12 End of General Tests

End of section.

3.1 Measurement of the ground wire

3.1.1 Measurement of the system ground wire (every 2 years)

Required time: 30 min

Required tools: Protective conductor meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure, and check that the ground wires are connected correctly.

Before performing the protective conductor test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.



WARNING

The ground wire meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective conductor meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

- » Perform protective conductor tests using measurement cables of a sufficient length or after the magnet has been ramped down.

- » The protective conductor resistance must not exceed **0.3 ohms**.

- Use the **ground wire meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI EPC MR electronic cabinet - ground bolt - line distribution box (limit value $\leq 0.3 \Omega$)

SI RF-filter plate - ground bolt (limit value $\leq 0.3 \Omega$)

SI Magnet cold head compressor - housing (limit value $\leq 0.3 \Omega$)

SI Option SEP - metal frame (limit value $\leq 0.3 \Omega$)

SI Option IFP - metal frame (limit value $\leq 0.3 \Omega$)

SI MRC (MRAWP) - Host - housing screw (limit value $\leq 0.3 \Omega$)

SI MRC (MRAWP) - TFT monitor - housing screw at the bottom (limit value $\leq 0.3 \Omega$)

SI Multiple socket for the console components (Line Distribution 1 – UPS) - ground screw (limit value $\leq 0.3 \Omega$)

SI Multiple socket for the console components (Line Distribution 2 – NO UPS) ground screw (limit value $\leq 0.3 \Omega$)

SI RF components on the magnet - RFCEL housing (limit value $\leq 0.3 \Omega$)

SI RF components on the magnet - TX-Box housing (limit value $\leq 0.3 \Omega$)

SI Magnet - magnet housing (limit value $\leq 0.3 \Omega$)

- All other options such as patient monitor, in-room MRC, Digicam, etc.



Ground wire measurement for CCD-Vario-Camera of PAMO5 is not necessary as it is supplied with low voltage.

SI Option 1 (limit value $\leq 0.3 \Omega$)

SI Option 2 (limit value $\leq 0.3 \Omega$)



Enter N.A. in the measured value field if the component/option is not present.

- Enter the values measured for each component in the protocol.



WARNING

Exceeding threshold values!

In case of electrical fault, exceeding a threshold value may result in death or serious injury.

- » Test to determine what exceeds the threshold value. Implement actions immediately to reduce the value prior to continuing work.

NOTICE**Option: In-room MRC**

Do not touch the trackball with the probe of the ground wire meter.

The high current used for measuring the resistance may damage the small contact springs of the trackball during the measurement.

- » To perform the measurement, use the uncoated screws at the back of the top cover of the In-Room MRC.

3.1.2 Measurement of the fixed Patient Table Ground Wire (every 2 years)

Required time: 5 min

Required tools: Protective conductor meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure and check that the ground wires are connected correctly.

Before performing the protective conductor test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

**WARNING**

The protective conductor meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective conductor meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

- » Perform protective conductor tests using measurement cables of sufficient length or after the magnet has been ramped down.

- Use the **protective conductor meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:

SI Fixed patient table - metal frame (limit value $\leq 0.3 \Omega$)

- » The protective conductor resistance must not exceed **0.3 ohms**.



Enter n.a. in the measured value field if the component is not present.

- Enter the values measured for each component in the protocol.

**WARNING**

Exceeding threshold values!

In case of electrical faults, exceeded threshold values may result in death or serious injury.

- » Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.

3.1.3 Measurement of the Dockable Patient Table Ground Wire (annually)

Required time: 5 min

Required tools: Protective conductor meter

Before starting the test, switch off the entire system. Ensure that it cannot be switched on inadvertently during the procedure and check that the ground wires are connected correctly.

Before performing the protective conductor test, read the operator manual for your measurement device. For some measurement devices, the resistance of the measurement cable must be subtracted from the measured results.

**WARNING**

The protective conductor meter is ferromagnetic. Do not bring it into the examination room while the magnet is at field.

Failure to comply may cause the protective conductor meter to become a projectile due to the magnetic force, resulting in serious injury or even death.

- » Perform protective conductor tests using measurement cables of sufficient length or after the magnet has been ramped down.

- Use the **protective conductor meter** and measure the **ohmic resistance** between the ground wire bar in the on-site power distribution cabinet and a non-insulated metallic part of the:
 - SI** Dockable patient table - metal frame (limit value $\leq 0.3 \Omega$)
 - SI** Option Combi Dockable Table - ground wire connection near floor dock pedal – right hand side (limit value $\leq 0.3 \Omega$)
 - SI** Option Combi Dockable Table - ground wire connection near floor dock pedal – left hand side (limit value $\leq 0.3 \Omega$)
- » The protective conductor resistance must not exceed **0.3 ohms**.
- Enter the values measured for each component in the safety-related Tests Protocol.

**WARNING**

Exceeding threshold values!

In case of electrical faults, exceeded threshold values may result in death or serious injury.

- » Test to determine what is currently exceeding the threshold value. Implement actions immediately to reduce the value prior to continuing work.

3.1.4 Measurement of the MRSC (MRWP) ground wire (annually)

Required time: 10 min

Required tools: Protective conductor meter

Use the ground wire meter to measure the ohmic resistance between the ground pin at the plug of the power distribution for the MRSC components and a non-insulated metallic part of the:

SI MRSC (MRWP) Host - housing screw (limit value $\leq 0.3 \Omega$)

SI MRSC (MRWP) TFT monitor - housing screw at the bottom (limit value $\leq 0.3 \Omega$)

SI Option 1 (limit value $\leq 0.3 \Omega$)

- » The protective conductor resistance must not exceed **0.3 ohms**.
- Enter the values measured for each component in the protocol.



Enter N.A. in the measured value field if the component/option is not present.

3.2 End of Electrical Tests

End of section.

4.1 Emergency power off switches

Required time: 15 min

Required tools: n.a.

SI Checking the emergency power off switches EPO (if installed)

Prerequisites:

The system contactor is switched on.

The system is not booted (system shutdown or press **F2** during boot).

- Press the emergency power off switch (EPO).
- Check that the system contactor in the electronics cabinet is de-energized.
- Check that all system components are voltage-free (e.g. imager, host computer).
- If a UPS (Uninterruptible Power Supply) is installed, it must also be switched off by pressing the EPO.
- Check that the **"Line"** LED at the alarm box is off.
- If there is an audible signal, press the **Alarm Silence** button on the alarm box to stop the signal.
 - » If the RF-room has an electric or pneumatic door, it has to be possible to open it in the event of a power failure.
- Check whether the RF-room door can be opened when the system is "powered off."

Release the emergency power off switch and reactivate the system by switching on the main power.

- Perform this test on all emergency power off switches in the equipment room, control room, and the examination room.

4.2 Gradient Supervision

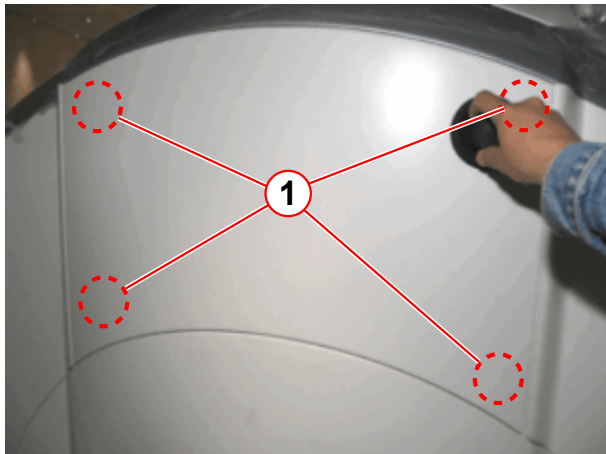
Required time: 40 min

Required tools: Smoke wick or test spray material number 10684091 , pocket lighter, screwdriver and 2.5 mm Allen key

Prerequisites: The display of the aspiration smoke detector shows OK (→ 1/ Fig. 58 Page 66). System is on.

- Remove the cover gradient cable.

Fig. 48: Gradient cable cover with suction cup



(1) Snap lock positions

- Attach the suction cup.
- Carefully pull out the cover gradient cable (plugged with snap locks).



WARNING

Accumulation of oxygen during helium filling.

The use of burning smoke wicks during helium filling could result in an open fire in the area of the magnet service turret.

» **Do NOT perform a smoke test during helium filling!**

SI Checking the gradient supervision

- Take the smoke wick out of the box.

Fig. 49: Smoke wick in the box



NOTICE**On-site smoke detectors****Avoid false alarm of the on-site smoke detector!**

- » Check if there is an on-site smoke detector. The smoke or test gas must not enter the on-site smoke detector.
- » Keep the smoke or test gas away from on-site smoke detectors.
- » Keep the smoke time as short as possible to minimize smoke concentration in the air.

- Light the tip of the smoke wick and allow it to burn for 10 - 15 seconds. Then blow out the flame.

Fig. 50: Burning smoke wick



(1) Plastic handle

- » The smoke wick provides a stable and continuous trail of smoke.

Fig. 51: Magnet service end - gradient connection



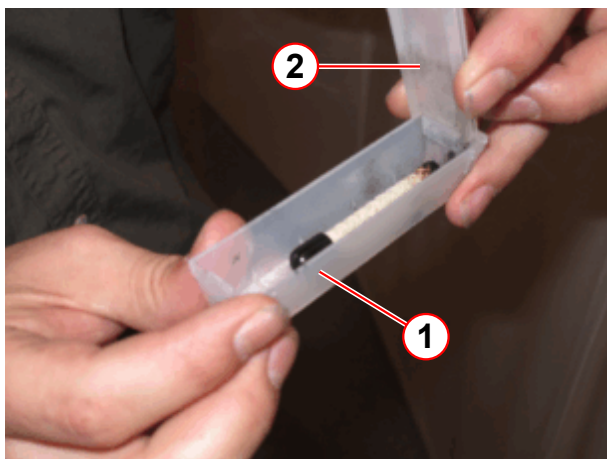
(1) Intake manifold

Fig. 52: Smoke test



- Hold the smoke wick behind the rear cover (→ Fig. 52 Page 61) for 5 seconds, so that the smoke is aspirated and transported to the detector.
- After use, immediately place the smoke wick into the empty box and close the lid carefully.

Fig. 53: Extinguishing the smoke wicks



- (1) Box
(2) Lid

» This will extinguish the smoke wick after approximately 15 seconds.



Alternatively, the test spray can be used. Spray the test gas dosed toward the intake manifold for 5 seconds. Dosed means in this context to press the spray button not completely. Distance between spray and intake manifold should be approximately 15-20 cm.

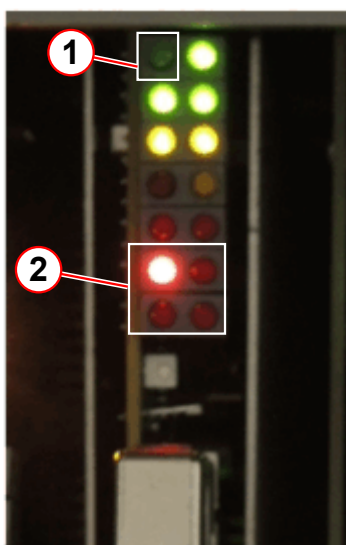
Fig. 54: Test spray



Expected test results:

- The smoke is being sucked (driven by 2 small fans) from the gradient manifold to the aspirating smoke detector. This will take up to 40 seconds (depending on the hose length), then the following results are expected simultaneously:
 - » The red "FIRE" lamp on the VLM device must light up.
 - » The gradient modulator in the GPA / GSSU is disabled (red LED's, error code 8).

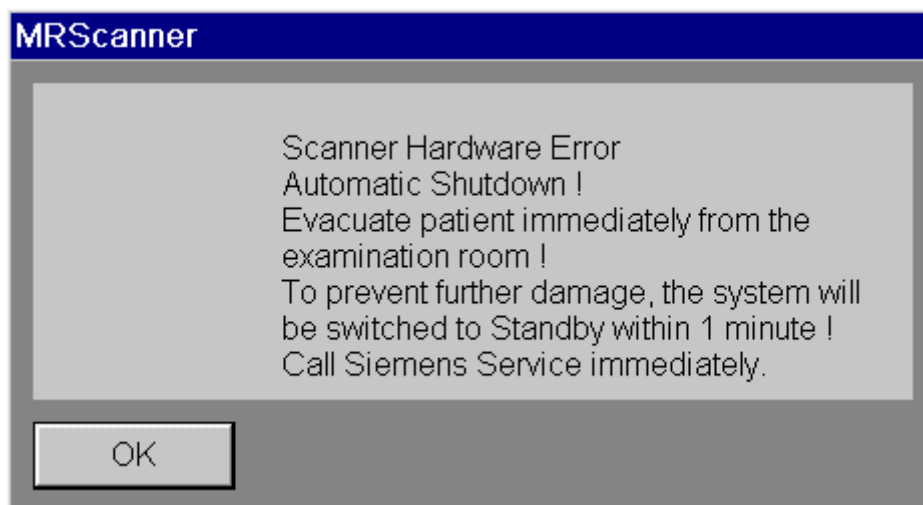
Fig. 55: Gradient modulator D40/ D41



- (1) Modulator off
- (2) Error code 8

- » An alarm message appears at the main console. (can also be checked in the error log).

Fig. 56: Pop-up "Automatic Shutdown"



and the scanner icon shows error .

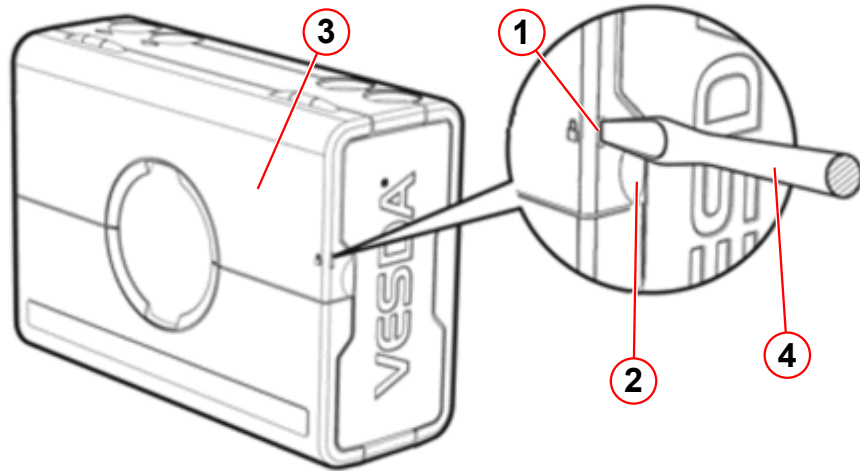
1 minute later,

- » The system must be switched to stand-by mode automatically.

Reset the system:

- Press a screwdriver into the security tab (→ 1/ Fig. 57 Page 65) to open the service access cover with the finger clip (→ 2/ Fig. 57 Page 65).

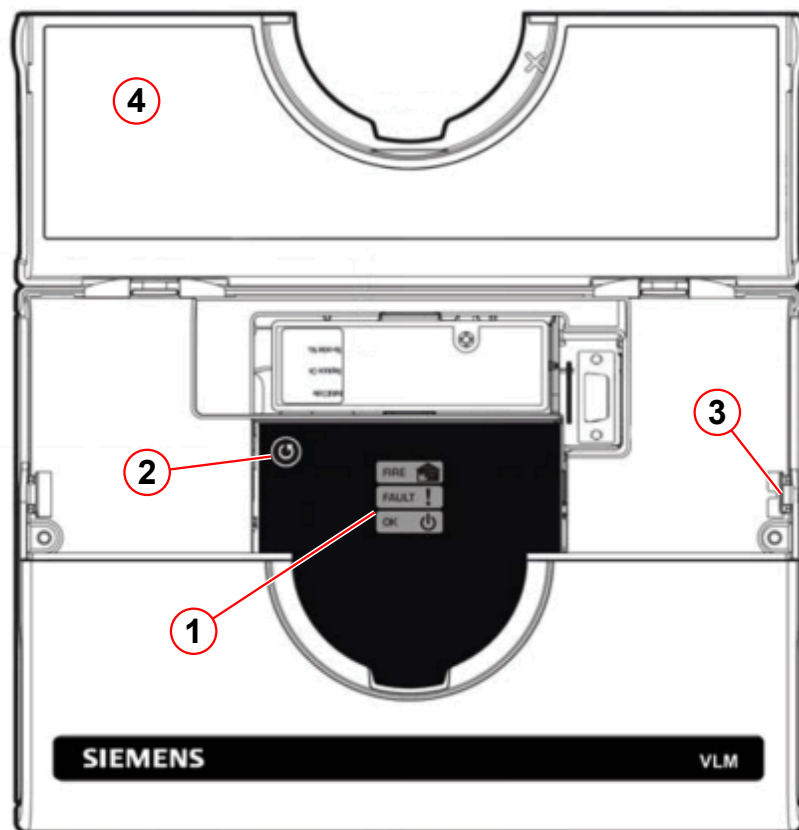
Fig. 57: Aspiration smoke detector ASD



- (1) Security tab
- (2) Finger clip
- (3) Service access cover
- (4) Screwdriver

- Reset the alarm by pressing the **reset button** (→ 2/Fig. 58 Page 66) for 2 seconds.

Fig. 58: Aspiration smoke detector



- (1) Status display
- (2) Reset button
- (3) Security tab
- (4) Service access cover



If the reset fails, the concentration of smoke inside the detector chamber is too high. Retry the reset after 1-2 minutes.

- » The aspiration smoke detector shows OK (→ 1/Fig. 58 Page 66).
- Start the system by pressing the **ON button** at the alarm box.

4.3 Patient table

4.3.1 STOP function

Required time: 5 min

Required tools: n.a.

Prerequisites: Table is in home position.

SI Checking the patient table stop function

- Press the **center move** button.
- While the table is moving toward the isocenter, press the **STOP** button at the right side of the patient table.
 - » The patient table stops; the Dot display indicates the table stop.
- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by turning the **STOP** button clockwise.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Press the center move button again to continue table movement.
- While the table is moving toward the isocenter, press the **STOP** button at the **left** side of the patient table
 - » The patient table stops; the display indicates the table stop.
- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by turning the **STOP** button clockwise.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Perform the patient table stop procedure with the **STOP** button for the **third control panel**, if present.

STOP button of the **intercom**

- Press the **center move** button on the control panel or on the console.

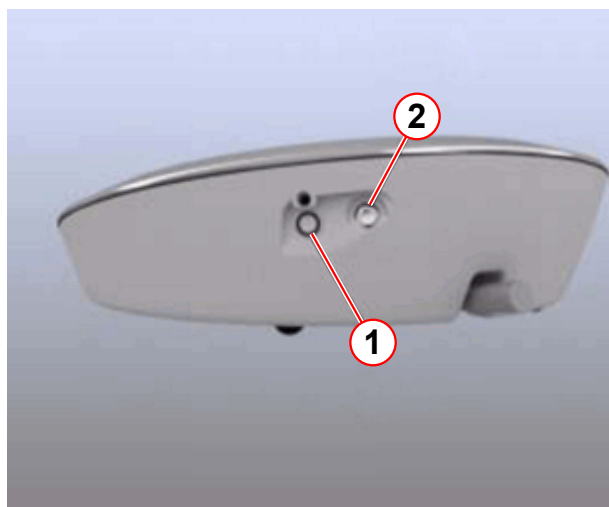
- While the table is moving toward the isocenter, press the **STOP** button of the intercom.

Fig. 59: Patient table stop on the intercom



- Make sure that table movement is not possible.
 - » On the MRC, the pop-up window opens < e.g. Sequence was stopped . . . , press **OK**
- Deactivate the table **STOP** button by pressing the **reset** button (→ 1/ Fig. 60 Page 68).

Fig. 60: Intercom



- (1) Reset patient table stop button
- (2) Audio connection

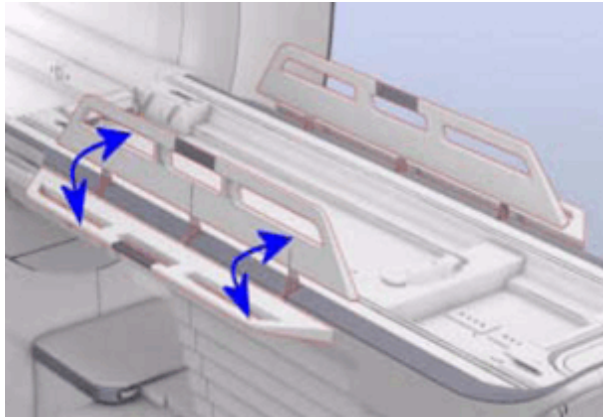
- Then press the up and down buttons simultaneously at the control panel magnet to reset the table.

Side railing (Dockable Table only)

- Press the **center move** button.

- While the table is moving toward the isocenter, lift up the side railing at the **left** side of the patient table

Fig. 61: Dockable table - side rails



- » The patient table stops; the Dot display indicates the table stop.
- Make sure that table movement is not possible.
- Move the side railing to the down position.
 - » Table movement is possible again
- Press the **center move** button.
- While the table is moving toward the isocenter, lift up the side railing at the **right** side of the patient table
 - » The patient table stops; the Dot display indicates the table stop.
- Make sure that table movement is not possible.
- Move the side railing to the down position.
 - » Table movement is possible again

4.3.2 Emergency release

Required time: 5 min

Required tools: n.a.

- SI** Checking the emergency release of the patient table
- Move the table into center position by pressing the **center move** button.

Fig. 62: Emergency release handle



- Pull the emergency release handle on the patient table.
- Slide the table all the way out.
 - » It must be possible to manually pull the table out of the magnet.

- Move the table manually approximately 5 mm and press the emergency release handle simultaneously until the emergency release handle moves to its original position.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Press the home button to move the table back into home position.

4.3.3 Bruising hazard on the service end

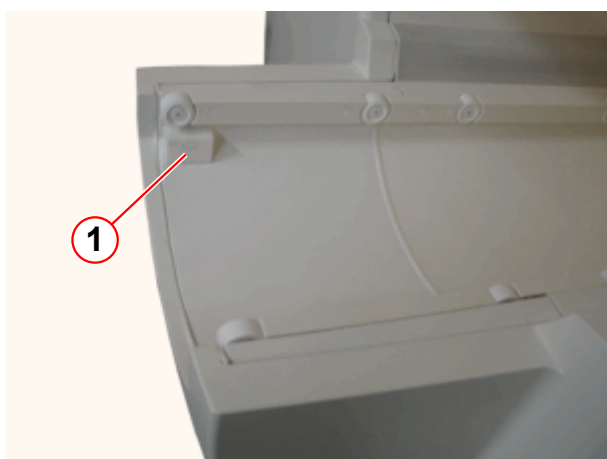
SI Checking the distance between the patient table and the wall in the RF-room

Required time: 5 min

Required tools: bruising hazard sign material number 3861718, tape material number 3474913; if required

- Check the mechanical end stop on the magnet bore (partial body) or on the rear support (whole body) for damage and correct mounting.

Fig. 63: Magnet service end - rear support

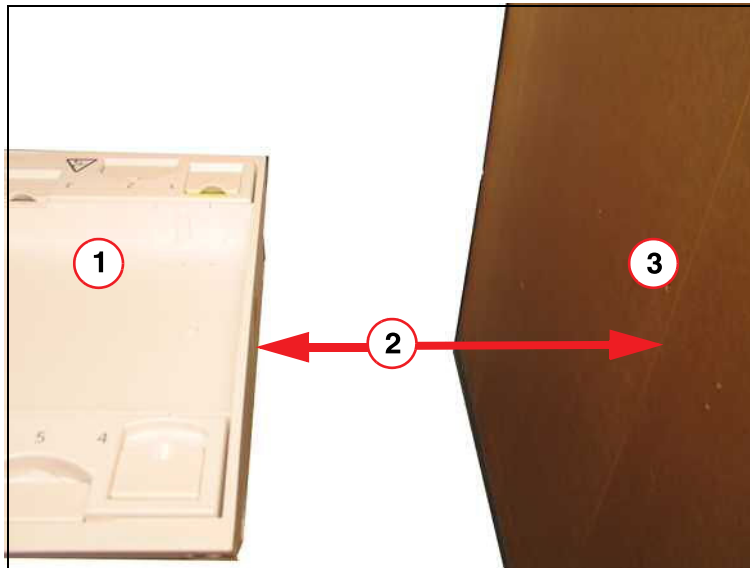


(1) Patient table - example of mechanical end stop

- Pull the emergency release handle on the patient table.

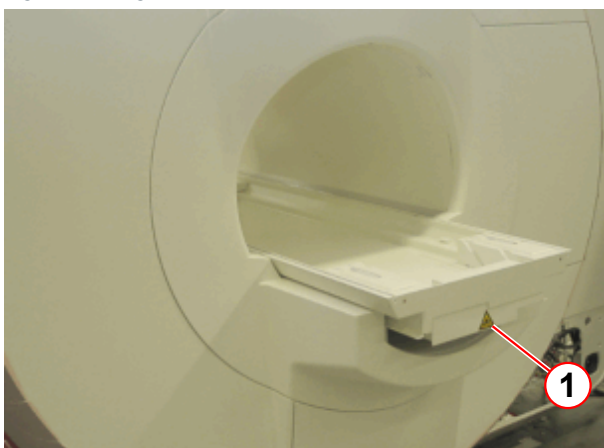
- Manually move the table all the way back (toward the magnet service end) until the table stops in the horizontal end position.
 - » If the distance between the patient table and the wall (service end) of the RF-room is **greater than** 500 mm, no further actions are necessary. Perform final steps.
 - » If the distance between the patient table and the wall (service end) of the RF-room is **less than** 500 mm, the bruising hazard sign has been posted at the rear of the patient table and the controlled access area has been protected with yellow and black tape .

Fig. 64: Bruising hazard - controlled access area



- (1) Patient table back in horizontal end position
- (2) Distance
- (3) Wall of the RF room

Fig. 65: Magnet service end



- (1) Sign for bruising hazards

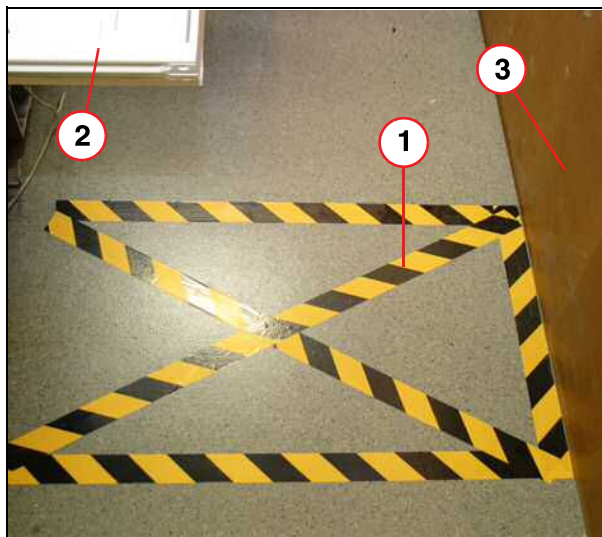
Fig. 66: Sign for bruising hazards



- Ensure that the bruising hazard sign has been posted at the rear of the patient table if the distance between the patient table and the wall (service end) is less than 500 mm.

- Install the missing sign immediately

Fig. 67: Service end area



- (1) Marking on the floor
- (2) Patient table
- (3) Wall of the RF room

- Ensure that the controlled access area has been protected with yellow and black tape if the distance between the patient table and the back wall is less than 500 mm.
- Install the missing tape immediately.

Final steps:

- Slide the table back into the magnet.
- Move the table manually approximately 5 mm and press the emergency release handle simultaneously until the emergency release handle moves to its original position.
- Confirm the pop-up windows on the MRC (MRAWP).
- To reset the table, press the IN and OUT buttons simultaneously on the control panel.
- Press the home button to move the table back into the home position.

4.3.4 Distance between the patient table and the cover

SI Checking the distance between the patient table and the cover

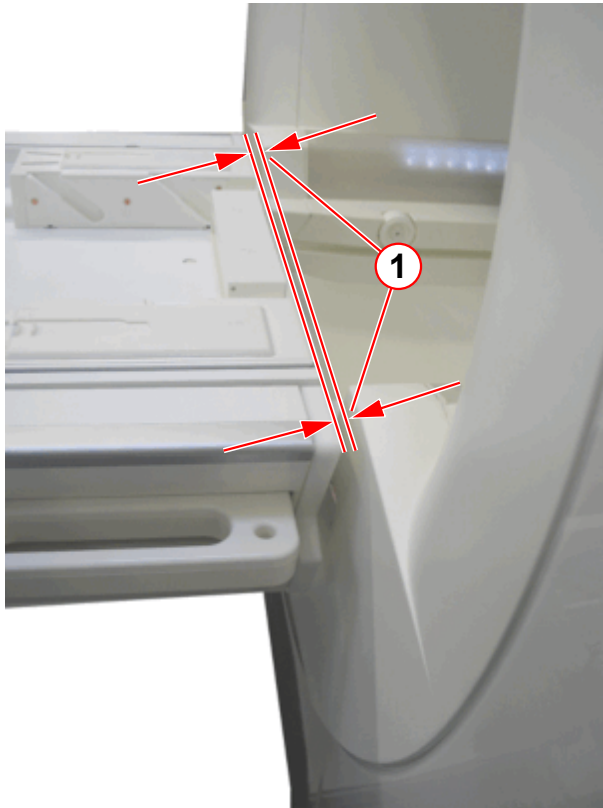
Required time: 5 min

Required tools: 8 mm Allen key or non-magnetic ruler

Prerequisites: Table is approximately 10 cm below home position.

1. Move the patient table up to the highest position.

Fig. 68: Patient table clearance



(1) Distance between patient table and cover

2. Measure the distance of the vertical clearance between the tabletop plate and the magnet cover.
 - » The distance between the tabletop plate and the magnet cover must be within the range of **< 8 mm**.
 - » If the distance is not within range, the patient table must be adjusted. Refer to Installation > Start-up > Check the mechanical adjustment of the patient table.

4.3.5 Table movement

Required time: 10 min

Required tools: Ruler

SI Checking the patient table movement

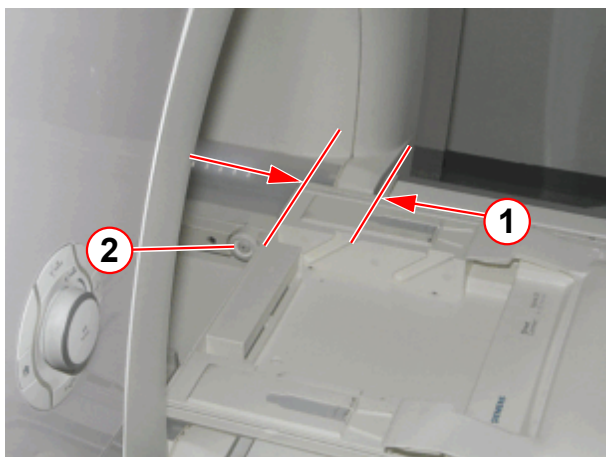
Vertical transfer position of the tabletop to the magnet bore.



When the tabletop moves from the home position into the magnet bore, it will be lifted slightly (approximately 1 mm) on the guide roller of the body coil. Check the vertical transfer position with the following procedure:

- Move the table about 10 cm into the magnet bore (→ 1/Fig. 69 Page 74). The tabletop is not on the guide roller (→ 2/Fig. 69 Page 74).
 - » Vertical transfer position to bore.

Fig. 69: Patient table



(1) Moving the tabletop into the bore (approx. 10 cm)

(2) Guide roller of the body coil

- Measure the distance between the upper side of the tabletop and the guide rail (→ 1/Fig. 71 Page 74).
 - » Write the distance down or mark the ruler. This is the value for the vertical transfer position.

Fig. 70: Tabletop is approx. 10 cm into the bore

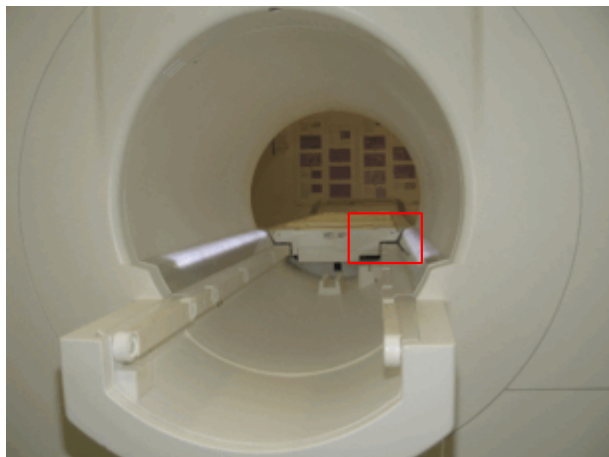
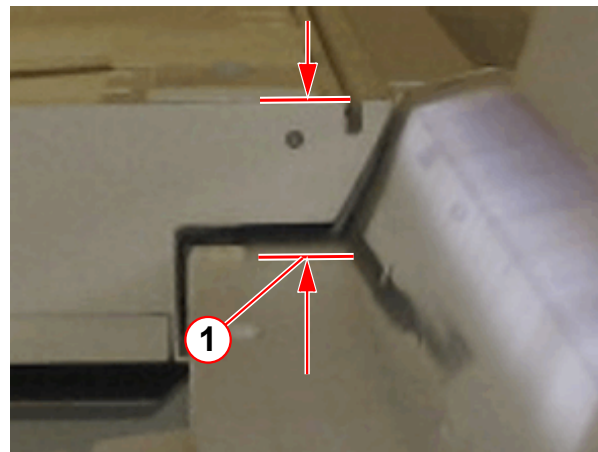


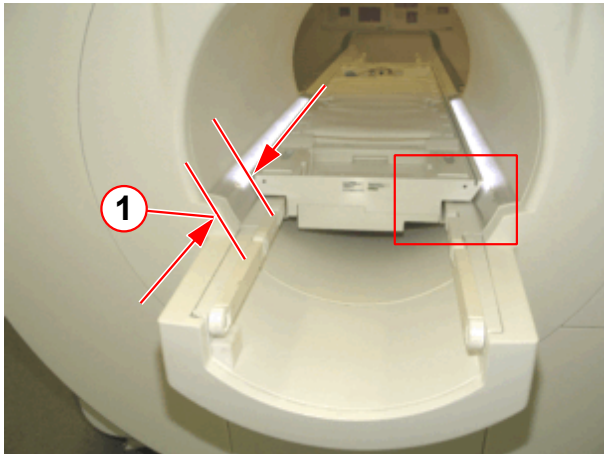
Fig. 71: Tabletop - slide rail



(1) Distance of upper side of tabletop to slide rail

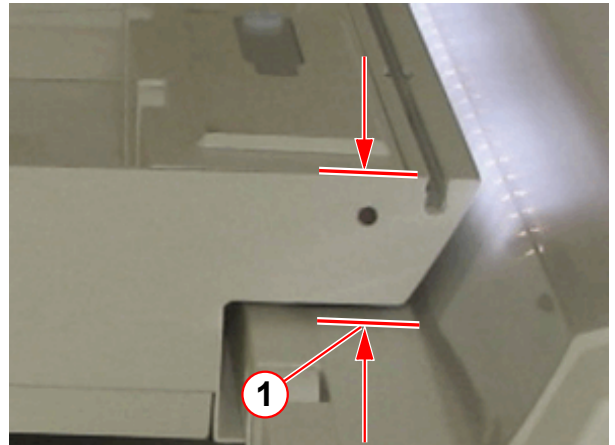
- Move the tabletop into the bore and stop the tabletop approximately 15 cm from the magnet cover (→ 1/Fig. 72 Page 75).
 - » The tabletop is on the guide roller of the body coil.

Fig. 72: Patient table



(1) Tabletop in the bore (approx. 15 cm from rear cover)

Fig. 73: Tabletop - slide rail



(1) Distance of upper side of tabletop to slide rail

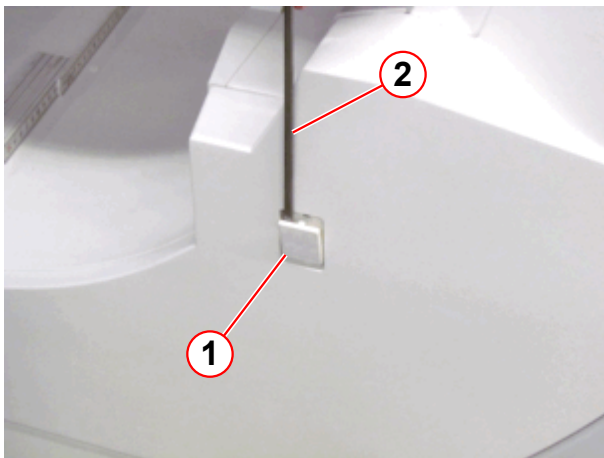
- Measure the distance between the upper side of the tabletop and the guide rail (→ 1/Fig. 73 Page 75). This is the reference distance of the tabletop in the bore.

» The reference distance of the tabletop in the bore must be $1 \text{ mm} \pm 0.5 \text{ mm}$ higher than the vertical transfer position distance.

If the distance is not within range, adjust the vertical transfer position as follows:

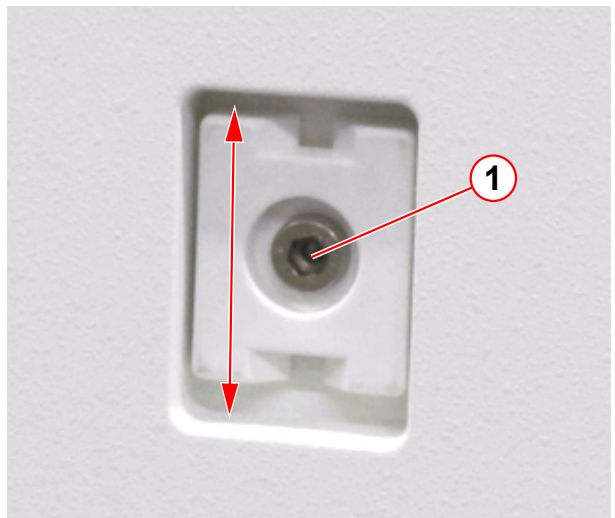
- Adjust the vertical transfer position

Fig. 74: Magnet cover - patient table reflector



(1) Reflector
(2) Screwdriver

Fig. 75: Reflector - retaining plate



(1) 3 mm Allen key

- Remove the reflector (→ 1/Fig. 74 Page 75) by using a screwdriver (→ 2/Fig. 74 Page 75)
- Loosen the screw (→ 1/Fig. 75 Page 75) and adjust the retaining plate.
 - » When moving the plate up, the vertical transfer position is higher.
 - » When moving the plate down, the vertical transfer position is lower.
- Fasten the screw (→ 1/Fig. 75 Page 75) and install the reflector (→ 1/Fig. 74 Page 75). Repeat the check.

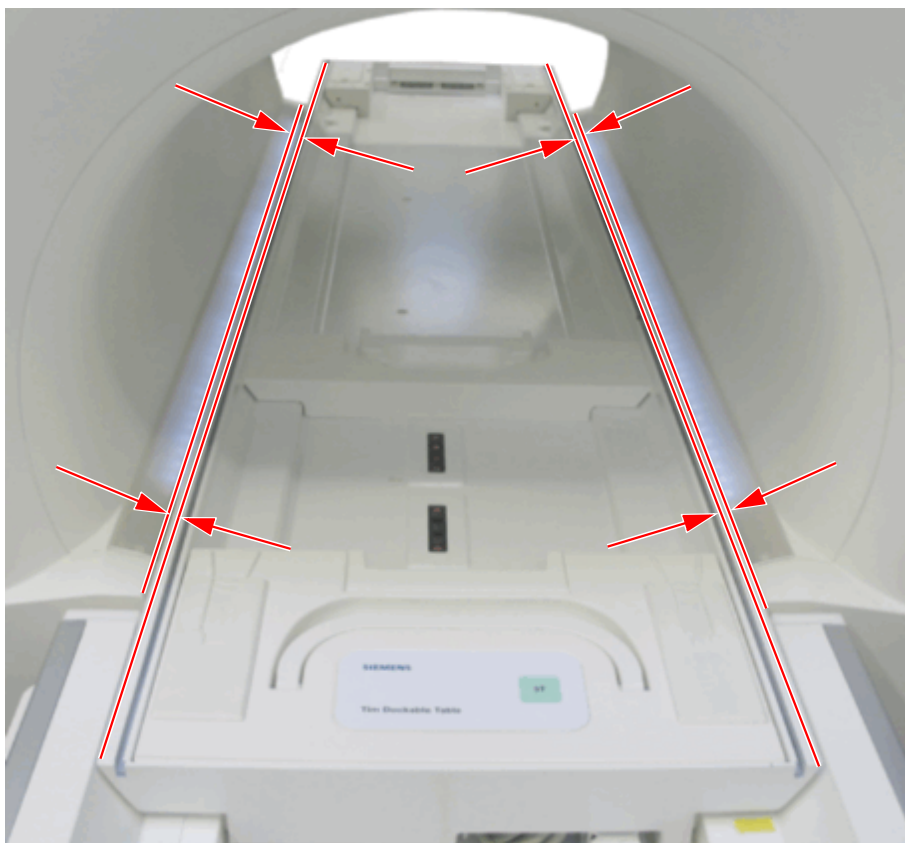


If more Dockable Tables are in use, check the transfer position of these tables too. If the transfer position is not OK adjust S20; refer to Replacement of Parts - Patient Handling - Patient Table

General movement

- Move the table all the way out.
 - » The table must move without any noticeable noise.
 - » The table stops in the horizontal end position.
- Lower the table all the way.
 - » The table must move without any noticeable noise.
 - » The table stops in the vertical end position.
- Move the table all the way up again.
 - » The table must move without any noticeable noise.
 - » The table stops in the vertical end position.
- Move the table into the magnet center.

Fig. 76: Patient table - parallelism



- » The patient table must move parallel to the body coil.

- Move the table all the way in.
 - » The table must move without any noticeable noise.
 - » The table stops in the horizontal end position.
- Move the table to the home position.

4.3.6 Squeeze bulb

Required time: 5 min

Required tools: n.a.

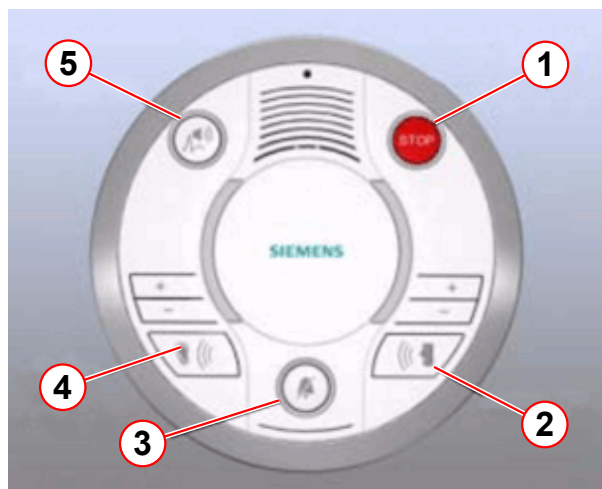
SI Checking the squeeze bulb

- Press the squeeze bulb on the patient table.

The following must occur:

- » Continuous tone over the intercom.
- » Flashing button on the intercom
- » The alarm can be reset by pressing the **Reset Alarm** or the **Talk** button on the intercom or on the small jog wheel at the control panel at the magnet.
- » You can speak with the patient by pressing the **Talk** button on the operating console. If necessary, adjust the volume using the +/- buttons.
- » Communication with the patient is established when you press the **Listen** button. If necessary, adjust the volume using the +/- buttons.

Fig. 77: Intercom,



- (1) Stop button
- (2) Talk button
- (3) Reset Alarm button
- (4) Listen button
- (5) Physio Signal button

4.3.7 Option dockable table: pressure activated switches

Required time: 5 min

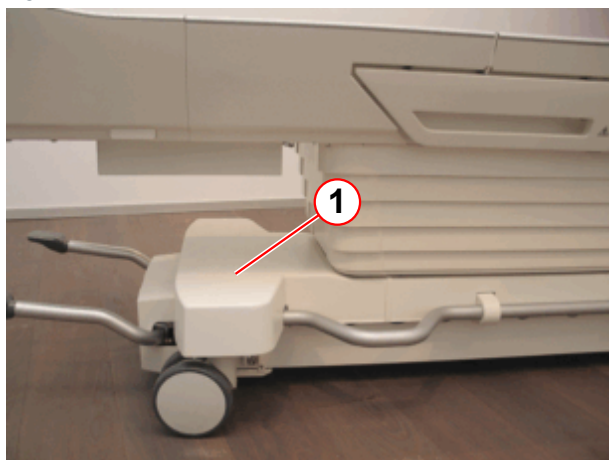
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

There are three pressure activated switches on the dockable table.

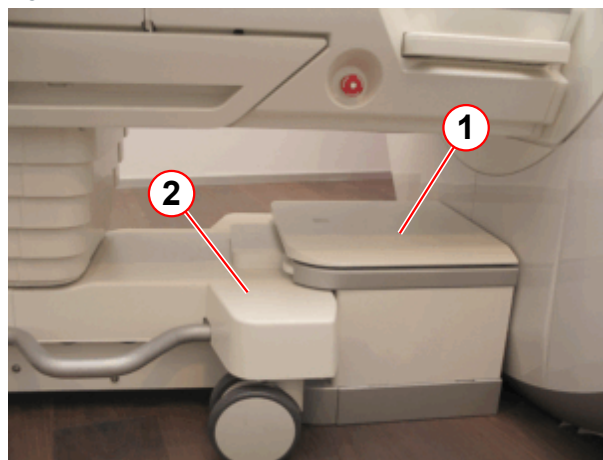
1. Under the chassis foot-end cover
2. Under the top cover of the docking station
3. Under the chassis head-end cover

Fig. 78: Dockable table



(1) Chassis foot-end cover

Fig. 79: Dockable table



(1) Top cover of the docking station
(2) Chassis head end cover

SI Checking the pressure activated switches of the option dockable table

- Move the table downward by pressing the down button
- During table movement, press the cover of the docking station at the top side to activate switch S60.
 - » The table stop movement
- Press the up button
 - » The table moves upward
- Press the down button again
- During table movement, press the chassis head-end cover downward to activate switch S60.
 - » The table stop movement
- Press the up button
 - » The table moves upward
- Press the down button again

- During table movement, press the chassis foot-end cover downward to activate switch S60.
 - » The table stop movement
- Press the up button
 - » The table moves upward

4.3.8 Option dockable table: emergency undock function

Required time: 10 min

Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

SI Checking the emergency undock function of the option dockable table

- Pull the emergency undocking handle.

Fig. 80: Emergency undock handle



- » The docking station releases the Dockable Table. The table is undocked.
- Move the Dockable Table away from the docking station.
- Move the Dockable Table into the docking station.
- Step on the undocking pedal.
 - » The emergency undock function will be repositioned.
- Step on the docking pedal.
 - » The Dockable Table is docked.
- Step on the docking pedal again.
 - » Stepping on the pedals has no effect on the table.

4.3.9 Option dockable table: chassis and wheels

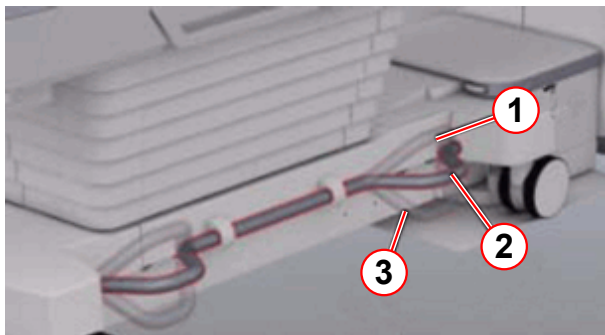
Required time: 5 min

Required tools: n.a.

SI Checking the chassis and the wheels (dockable table only)

- Undock the Dockable Table from the docking station.
- Move the table by a few meters.
 - » The table must move smoothly without any noticeable noise.

Fig. 81: Dockable table - lateral lever



- (1) 5th wheel is disabled
- (2) 5th wheel is activated
- (3) Table is braked

- Activate the 5th wheel by using the lateral lever (→ 2/Fig. 81 Page 80).
 - » The 5th wheel is activated.
- Move the table.
 - » The maneuverability is improved.
- Press the lateral lever (→ 3/Fig. 81 Page 80).
 - » It must not be possible to move the Dockable Table

Table top lock

- Move the table top in a horizontal direction.
 - » It is possible to move the table top approximately 10 mm. Then stop the table top.

4.3.10 Option dockable table: lowering level of the undocked table

Required time: 5 min

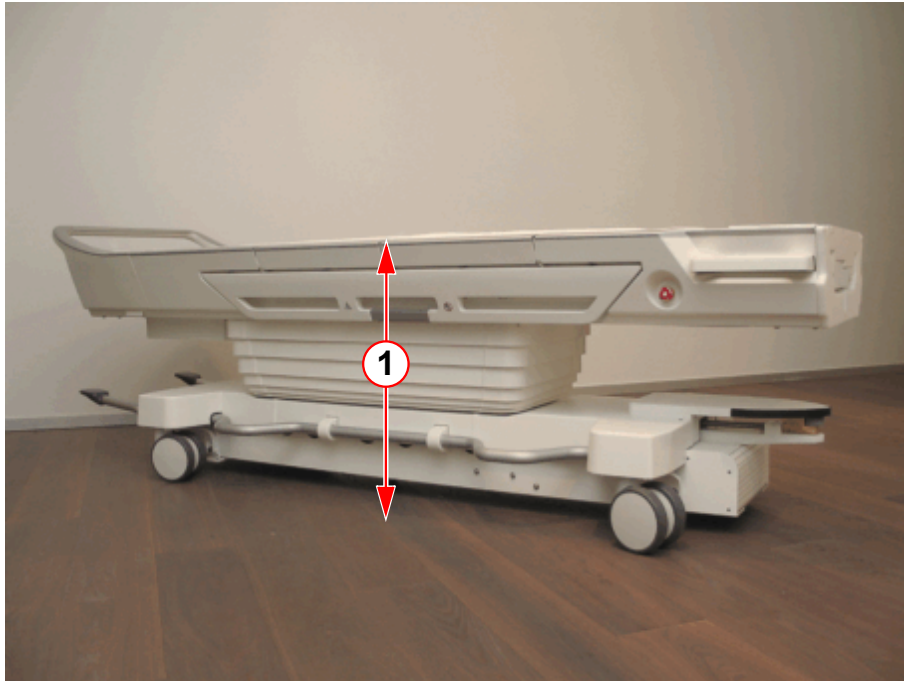
Required tools: n.a.

Prerequisites: The table is undocked.

SI Checking the lowering level of the undocked table (dockable table only)

- Lower the patient table all the way by stepping on the down pedal.
- Check the lowering level
 - » The distance between the tabletop plate and the floor must be 600 mm or greater.

Fig. 82: Dockable table



(1) Distance between tabletop and floor

4.4 ERDU

Required time: 15 min

Required tools: Screwdriver

SI Magnet Stop button test (ERDU)

4.4.1 Prerequisites



WARNING

Restrict Access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

Unauthorised personnel must be excluded from the imaging suite.

» Secure the room and prevent unauthorised access.

- Restrict access to the magnet if it is at field when the test is being performed.
- Check that all alarm and error indications show that the **system is OK**.
- Remove the appropriate covers to gain access to the **MSUP** and **Service Magnet Stop button**.

4.4.2 Testing the Magnet Stop button (ERDU)



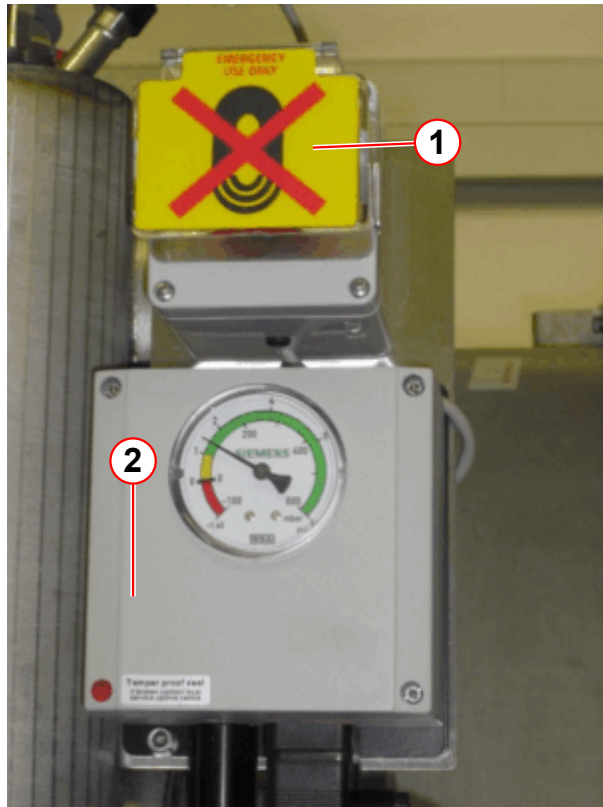
CAUTION

Magnet quench!

A magnet quench is caused by pressing the service magnet stop button (ERDU).

» Do not operate the Service Magnet Stop button unless an emergency magnet quench is required.

Fig. 83: Service Magnet Stop button



- (1) Service Magnet Stop button
- (2) Pressure gauge

- The Service Magnet Stop button is located above the pressure gauge.
- Check that the button is not activated.
- Test the Service Magnet Stop button.
 - » Press the button to ensure the MSUP audible alarm sounds
 - » Reset the button - The alarm will stop.

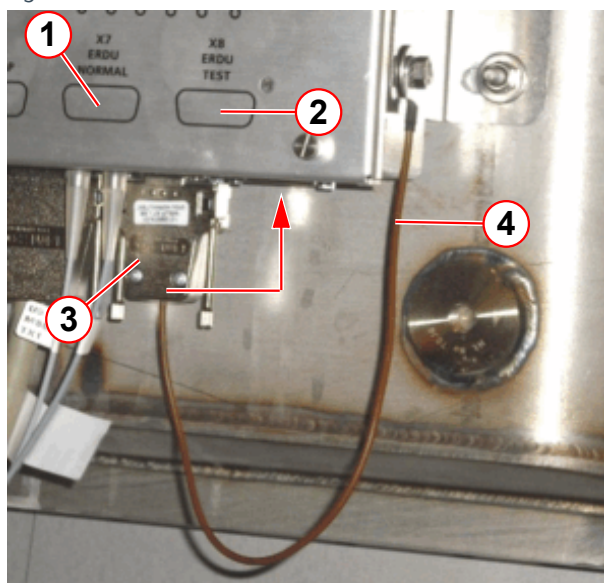


When the ERDU test connector is disconnected from BOTH X7 and X8, IT IS NOT POSSIBLE TO QUENCH THE MAGNET.

When the ERDU test connector is disconnected from the MSUP (X7), the customer site Magnet Stop button WILL NOT Quench the magnet. The MSUP audible warning is activated.

When ERDU test connector is connected to (X8), the MSUP audible warning continues. The Service Magnet Stop button is now active.

Fig. 84: ERDU test connector



- (1) ERDU NORMAL
- (2) ERDU TEST
- (3) ERDU test connector
- (4) ERDU connector strap

The ERDU test connector is fitted to **X7** during normal operation.

- Disconnect the test connector from **X7** (ERDU NORMAL).
 - » MSUP audible alarm is activated.
- Connect it to **X8** (ERDU TEST).
 - » On the alarm box, check the audible alarm sounds.
 - » Press AUDIO ALARM OFF to silence the alarm.
(→ 5/ Fig. 85 Page 85)
- Check that the ERDU TEST LEDs are NOT illuminated.

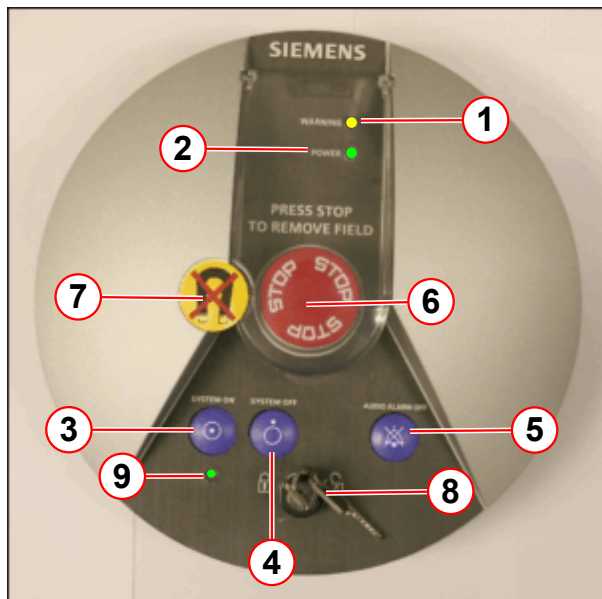


If the MSUP audible warning is not sounding, DO NOT CONTINUE.
Investigate the cause of the problem.



The Magnet Stop button locks when depressed.
Turn the button clockwise to unlock.

Fig. 85: Alarm box - LEDs and buttons

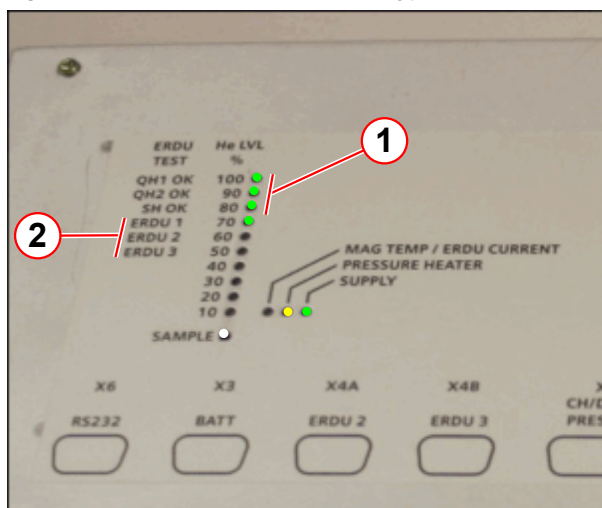


- (1) LED WARNING = yellow
- (2) LED POWER = green
- (3) Button - SYSTEM ON
- (4) Button - SYSTEM OFF
- (5) Button - AUDIO ALARM OFF
- (6) Button - MAGNET STOP (ERDU)
- (7) Tamper-proof seal
- (8) Power switch locking key
- (9) SYSTEM ON LED = green

At the alarm box:

- Remove the tamper-proof seal (if required).
- Press the Magnet Stop button.
 - » On the alarm box, check the audible alarm sounds and the yellow warning LED flashes.
 - » Press AUDIO ALARM OFF to silence the alarm.

Fig. 86: ERDU test - LED indicators (typical)



- (1) QH1, QH2, SH = green LED (all ON)
- (2) ERDU 1,2,3 = green LEDs (ON only when tested)

At the MSUP:

- Check QH1, QH2, SH LEDs are illuminated.
- Check ERDU 1 LED is illuminated.
- Reset the Magnet Stop button on the alarm box.
- Check the MSUP LEDs return to normal.
 - » QH1, QH2, SH and ERDU 1 are NOT illuminated.

In the **Magnet Room**:

- Press the **Magnet Stop** button.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.

- Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 2 LED** is illuminated.
- **Reset the Magnet Stop button** in the Magnet Room.
- **Check** the LEDs return to **normal**.

An optional **3rd Magnet Stop button** may be fitted. Follow the steps below:

- Press the **Magnet Stop button**.
 - Check the **audible alarm** sounds, and the **warning LED** flashes on the **alarm box**.
 - Press **AUDIO ALARM OFF** to silence the alarm.
- At the **MSUP**:
 - Check **QH1, QH2, SH LEDs** are illuminated.
 - Check **ERDU 3 LED** is illuminated.
- **Reset the 3rd Magnet Stop button**.
- **Check** the LEDs return to **normal**.

4.4.3 Final steps

- Check **ALL Magnet Stop buttons** are **not activated**.
- **Disconnect** the ERDU test connector **from X8** (ERDU TEST) and reconnect it to **X7** (ERDU NORMAL).
 - » On the **alarm box**, check the **audible alarm** sounds.
- **At the MSUP**:
 - » Check the **MSUP audible alarm** is **deactivated**.
 - » Check the **MSUP LEDs** return to **normal**.
- **At the alarm box**:
 - » Press **AUDIO ALARM OFF** to silence the alarm.
- **Check** the **warning LED** is not illuminated '**ON**'.



If the LED is on, or flashing, the fault must be checked.
Do not disconnect the ERDU battery.

4.4.4 ERDU Test status

- Access the **Service Software** at the **MRC**.

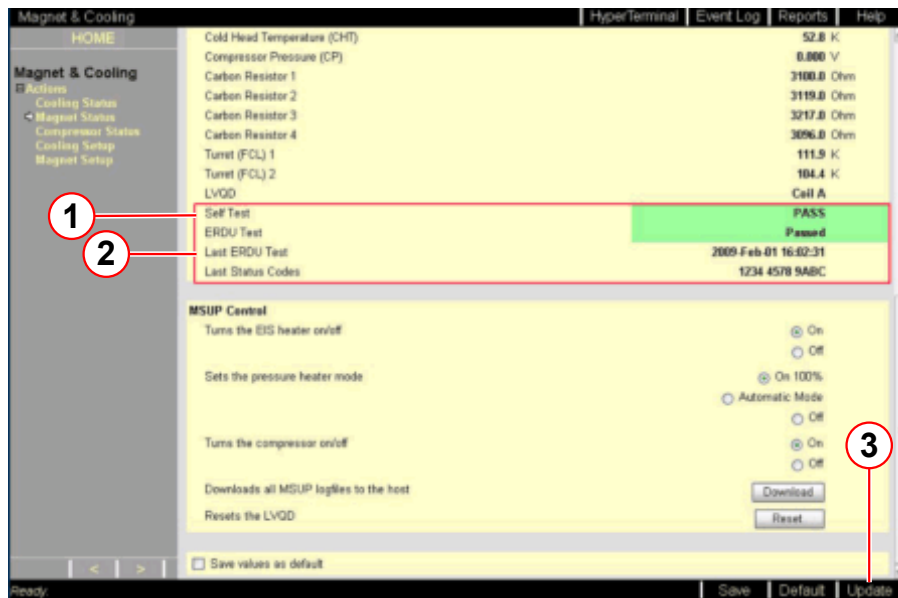
- Select the menu item **Magnet & Cooling - Magnet Status**.



Data is only accurate from the last refresh of the page, or from when the page was opened last.

- Check the status of the **ERDU Test** (→ 1/Fig. 87 Page 87) and the date of the **Last ERDU Test**. (→ 2/Fig. 87 Page 87)

Fig. 87: ERDU Test status



- (1) ERDU Test
- (2) Last ERDU Test
- (3) Update

4.5 Combi Dockable Table

4.5.1 Option Combi Dockable Table: STOP Function

Required time: 5 min

Required tools: n.a.

Prerequisites: Combi Dockable Table is docked to MR, table is in home position.

SI Option Combi Dockable Table: Checking the patient table stop function

- Press the **center move** button.
- While the table is moving toward the isocenter, press the **STOP** button at the right side of the patient table.
 - » The patient table stops; the Dot display indicates the table stop.
- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by turning the **STOP** button clockwise.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Press the center move button again to continue table movement.
- While the table is moving toward the isocenter, press the **STOP** button at the **left** side of the patient table
 - » The patient table stops; the display indicates the table stop.
- Make sure that table movement is not possible.
- Deactivate the table **STOP** button by turning the **STOP** button clockwise.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Perform the patient table stop procedure with the **STOP** button for the **third control panel**, if present.

STOP button of the **intercom**

- Press the **center move** button on the control panel or on the console.

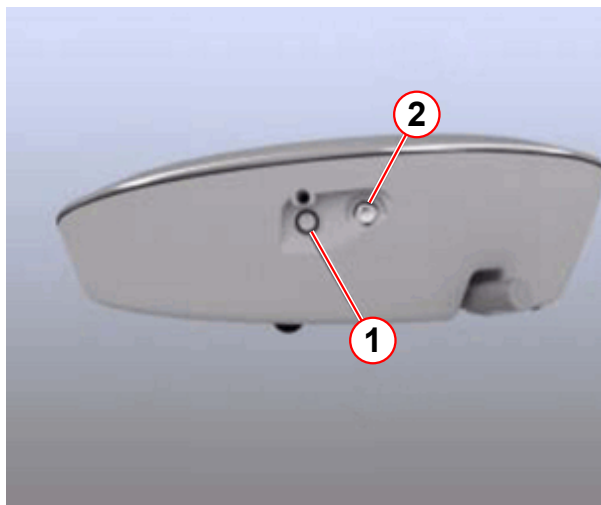
- While the table is moving toward the isocenter, press the **STOP** button of the intercom.

Fig. 88: Patient table stop on the intercom



- Make sure that table movement is not possible.
 - » On the MRC, the pop-up window opens < e.g. Sequence was stopped . . . , press **OK**
- Deactivate the table **STOP** button by pressing the **reset** button (→ 1/ Fig. 89 Page 89).

Fig. 89: Intercom



- (1) Reset patient table stop button
- (2) Audio connection
- Then press the up and down buttons simultaneously at the control panel magnet to reset the table.

4.5.2 Option Combi Dockable Table: Emergency Release

Required time: 5 min

Required tools: n.a.

SI Option Combi Dockable Table: Checking the emergency release

- Move the table into center position by pressing the **center move** button.

Fig. 90: Emergency release handle



- Pull the emergency release handle on the patient table.
- Slide the table all the way out.
 - » It must be possible to manually pull the table out of the magnet.

- Move the table manually approximately 5 mm and press the emergency release handle simultaneously until the emergency release handle moves to its original position.
- Then press the up and down buttons simultaneously at the control panel to reset the table.
- Press the home button to move the table back into home position.

4.5.3 Option Combi Dockable Table: Distance between the patient table and the cover

SI Option Combi Dockable Table: Checking the distance between the patient table and the cover

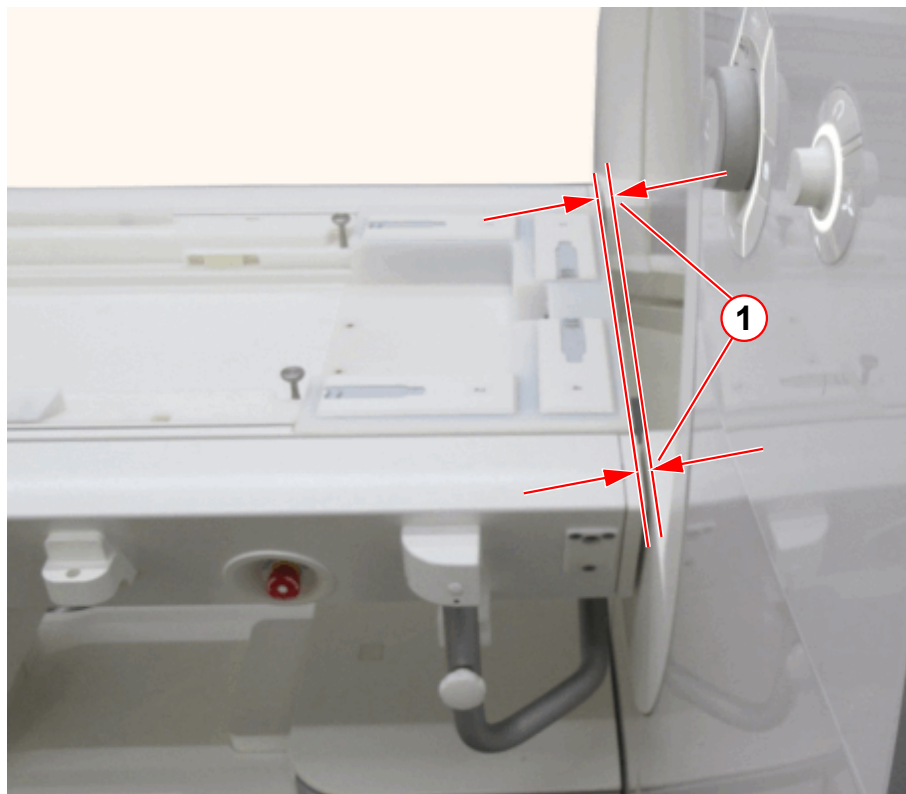
Required time: 5 min

Required tools: 8 mm non-magnetic Allen key or non-magnetic ruler

Prerequisites: Table is approximately 10 cm below home position.

1. Move the patient table up to the highest position.

Fig. 91: Patient table clearance



(1) Distance between patient table and cover

2. Measure the distance of the vertical clearance between the tabletop plate and the magnet cover (→ 1/ Fig. 91 Page 91).
 - » The distance between the tabletop plate and the magnet cover must be within the range of **< 8 mm**.
 - » If the distance is not within range, the patient table must be adjusted. Refer to Installation > Installation Instructions > Check the mechanical adjustment of the patient table.

4.5.4 Option Combi Dockable Table: Table movement

Required time: 15 min

Required tools: Ruler

SI Option Combi Dockable Table: Checking the patient table movement

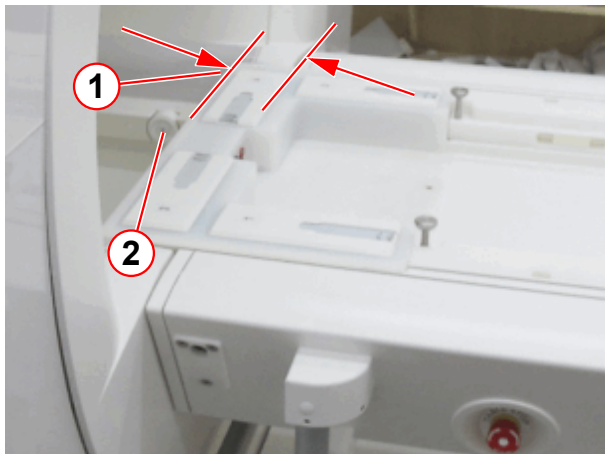
Vertical transfer position of the tabletop to the magnet bore.



When the tabletop moves from the home position into the magnet bore, it will be lifted slightly (approximately 1 mm) on the guide roller of the body coil. Check the vertical transfer position with the following procedure:

- Move the table about 10 cm into the magnet bore (→ 1/Fig. 92 Page 92). The tabletop is not on the guide roller (→ 2/Fig. 92 Page 92).
 - » Vertical transfer position to bore.

Fig. 92: Combi Dockable Table



(1) Moving the tabletop into the bore (approx. 10 cm)

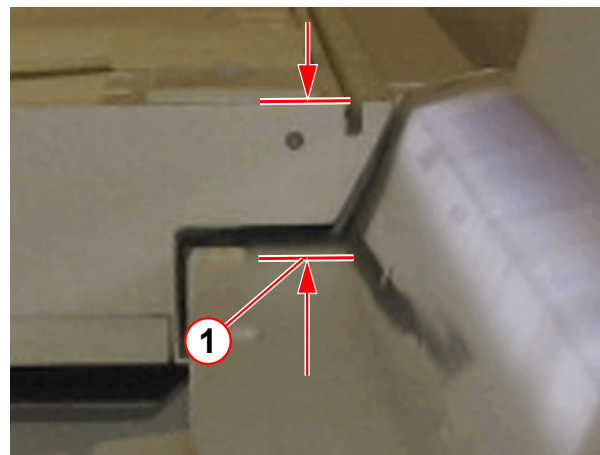
(2) Guide roller of the body coil

- Measure the distance between the upper side of the tabletop and the slide rail (→ 1/Fig. 94 Page 92).
 - » Write the distance down or mark the ruler. This is the value for the vertical transfer position.

Fig. 93: Tabletop is approx. 10 cm inside the bore



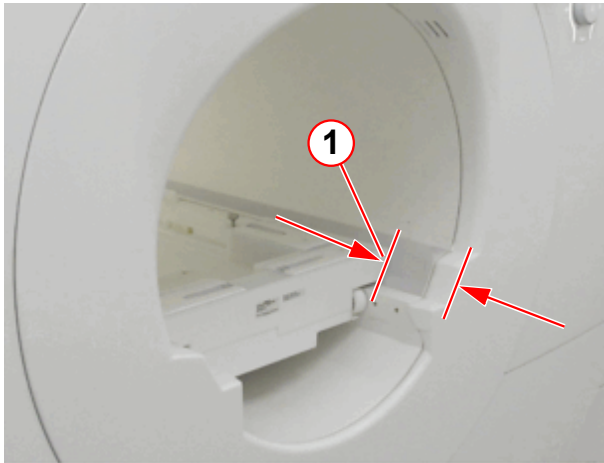
Fig. 94: Tabletop - slide rail



(1) Distance of upper side of tabletop to slide rail

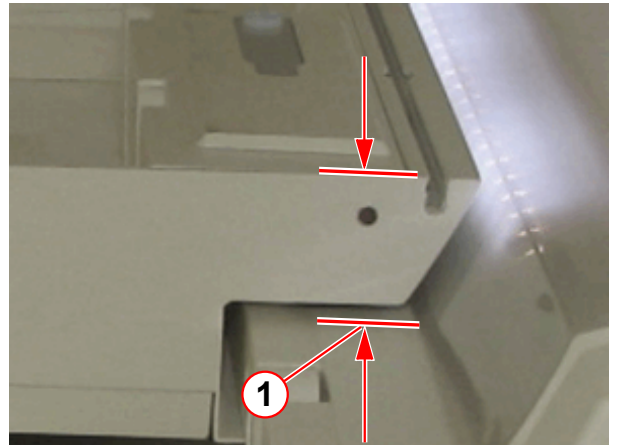
- Move the tabletop into the bore and stop the tabletop approximately 15 cm from the magnet cover (→ 1/Fig. 95 Page 93).
 - » The tabletop is on the guide roller of the body coil.

Fig. 95: Combi Dockable Table



(1) Tabletop in the bore (approx. 15 cm from rear cover)

Fig. 96: Tabletop - slide rail



(1) Distance of upper side of tabletop to slide rail

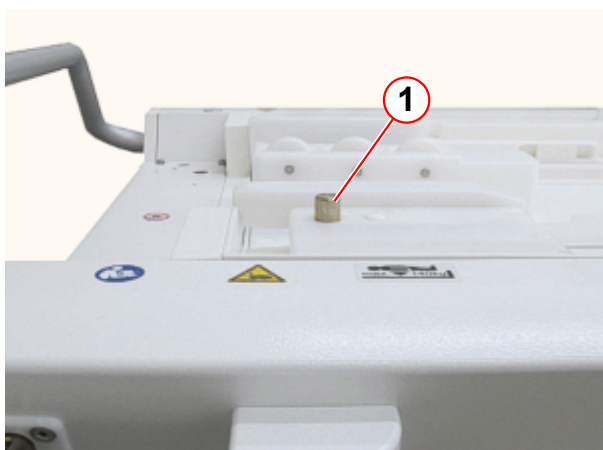
- Measure the distance between the upper side of the tabletop and the slide rail (→ 1/ Fig. 96 Page 93). This is the reference distance of the tabletop in the bore.
 - » The reference distance of the tabletop in the bore must be $1 \text{ mm} \pm 0.5 \text{ mm}$ higher than the vertical transfer position distance.

If the distance is not within range, adjust the vertical transfer position (reflector or if more mobile tables are in use S20), refer to Replacement of Parts - Patient Handling - Patient Table.

General movement

- Move the table all the way out.
 - » The locking bolt of the tabletop is moved down (→ 1/ Fig. 97 Page 93).

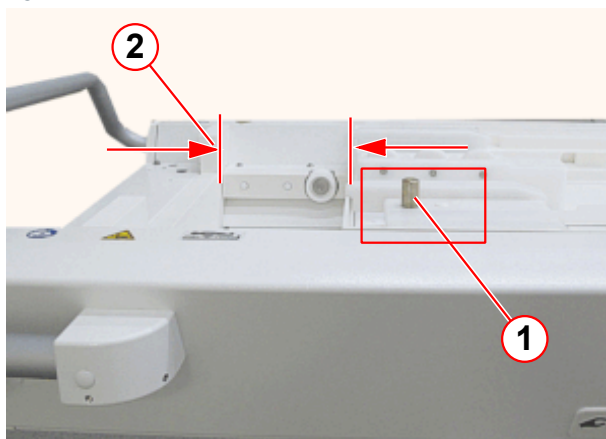
Fig. 97: Combi Dockable Table - tabletop is in home position



(1) Locking bolt of the tabletop (moved down)

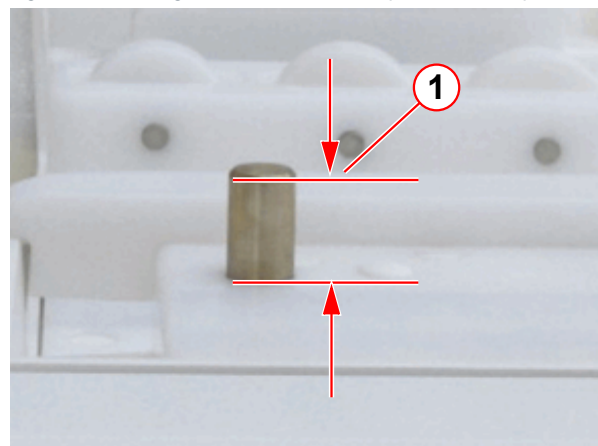
- Move the table approximately 30 cm into the magnet bore (→ 1/ Fig. 98 Page 94).

Fig. 98: Combi Dockable Table -



- (1) Locking bolt of the tabletop
 (2) Moving the tabletop into the magnet bore (approx. 30 cm)

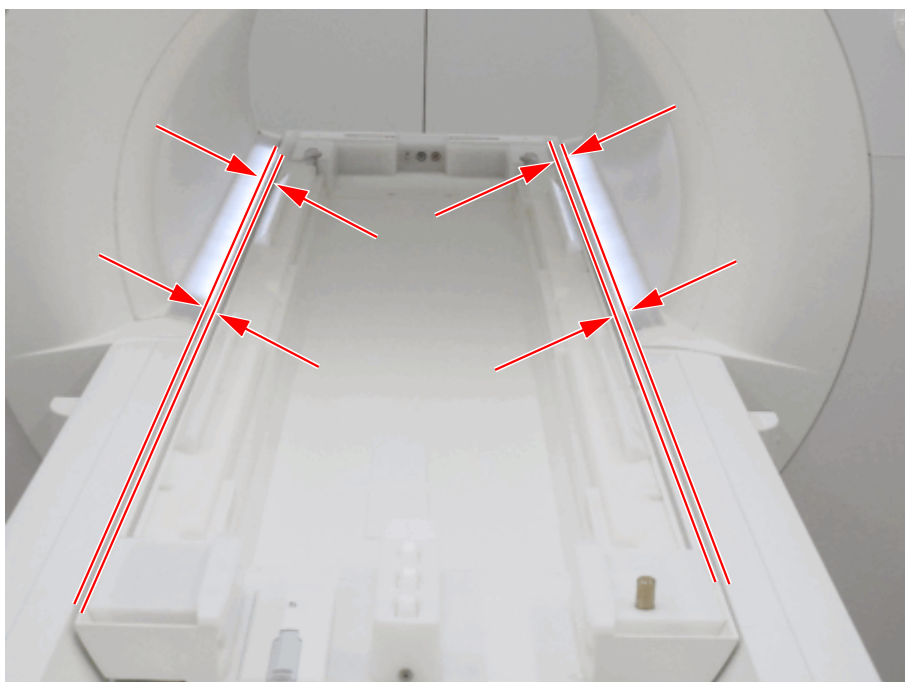
Fig. 99: Locking bolt of the table top is moved up



- (1) Distance of upper side of tabletop to the top of the locking bolt

- » The locking bolt of the tabletop moves up (→ 1/ Fig. 99 Page 94).
- Check the distance of the upper side of the tabletop to the top of the locking bolt. The distance depends on the tables and transfer boards used for the transfer.
 - » Distance Artis: ≥ 24 mm
 - » Distance TRUMPF: ≥ 24 mm
 - » Distance MAQUET: ≥ 39 mm
- Move the table into the magnet center.
 - » The patient table must move parallel to the body coil (→ Fig. 100 Page 94).

Fig. 100: Combi Dockable Table - parallelism



- Move the table all the way in.
 - » The table must move without any noticeable noise.
 - » The table stops in the horizontal end position.
- Move the table all the way out.
 - » The table must move without any noticeable noise.
 - » The table stops in the horizontal end position.
- Lower the table all the way.
 - » The table must move without any noticeable noise.
 - » The table stops in the vertical end position.
- Move the table all the way up again.
 - » The table must move without any noticeable noise.
 - » The table stops in the vertical end position.

4.5.5 Option Combi Dockable Table: Squeeze bulb

Required time: 5 min

Required tools:n.a.

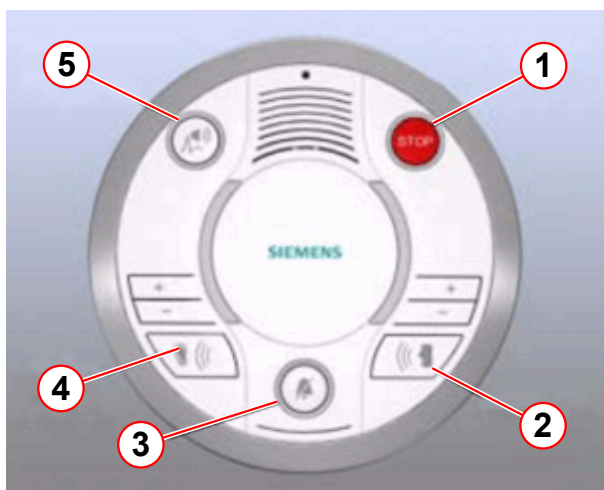
SI Option Combi Dockable Table: Checking the squeeze bulb

- Press the squeeze bulb on the patient table.

The following must occur:

- » Continuous tone over the intercom.
- » Flashing button on the intercom
- » The alarm can be reset by pressing the **Reset Alarm** or the **Talk** button on the intercom or on the small jog wheel at the control panel at the magnet.
- » You can speak with the patient by pressing the **Talk** button on the operating console. If necessary, adjust the volume using the +/- buttons.
- » Communication with the patient is established when you press the **Listen** button. If necessary, adjust the volume using the +/- buttons.

Fig. 101: Intercom,



- (1) Stop button
- (2) Talk button
- (3) Reset Alarm button
- (4) Listen button
- (5) Physio Signal button

4.5.6 Option Combi Dockable Table: Pressure activated switches

Required time: 5 min

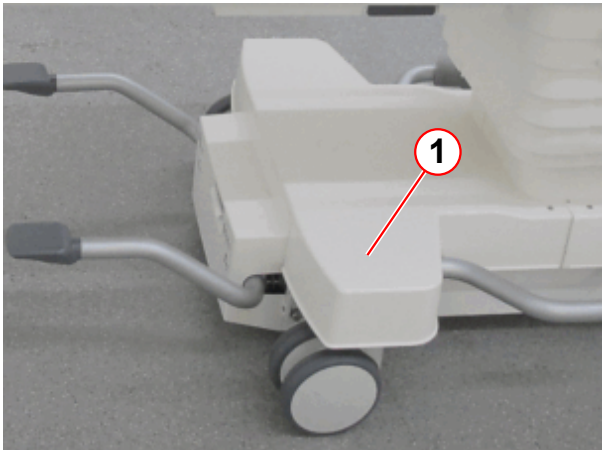
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

There are three pressure activated switches on the dockable table.

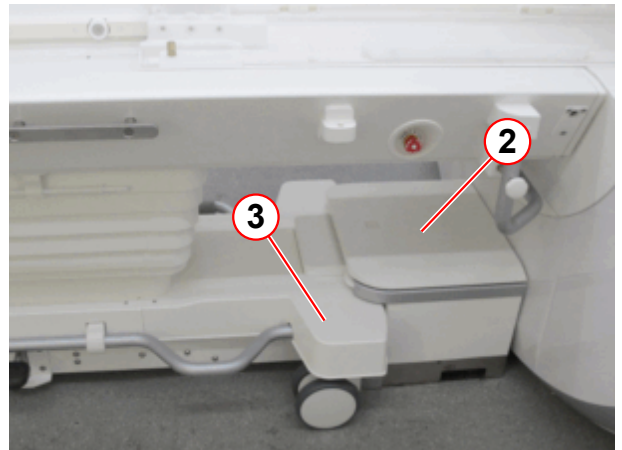
1. Under the chassis foot-end cover
2. Under the top cover of the docking station
3. Under the chassis head-end cover

Fig. 102: Combi Dockable Table



(1) Chassis foot-end cover

Fig. 103: Combi Dockable Table

(2) Top cover of the docking station
(3) Chassis head end cover

SI Option Combi Dockable Table: Checking the pressure activated switches

- Move the table downward by pressing the down button
- During table movement, press the cover of the docking station at the top side to activate switch S60 (→ 2/ Fig. 103 Page 97) .
 - » The table stop movement
- Press the up button
 - » The table moves upward
- Press the down button again
- During table movement, press the chassis head-end cover downward to activate switch S60 (→ 3/ Fig. 103 Page 97).
 - » The table stop movement
- Press the up button
 - » The table moves upward
- Press the down button again
- During table movement, press the chassis foot-end cover downward to activate switch S60 (→ 1/ Fig. 102 Page 97).
 - » The table stop movement
- Press the up button
 - » The table moves upward

4.5.7 Option Combi Dockable Table: Emergency Undock Function

Required time: 10 min

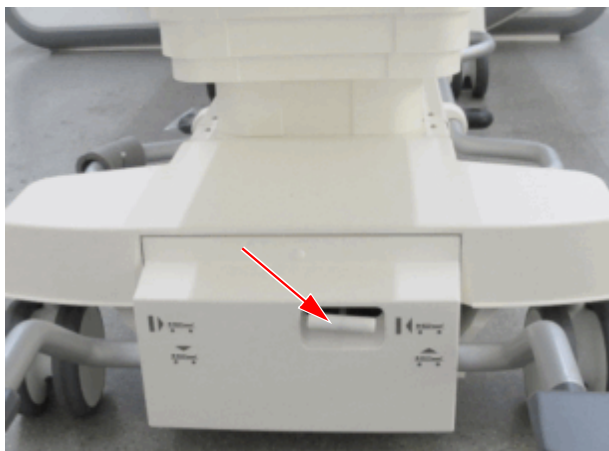
Required tools: n.a.

Prerequisites: Patient table is docked and in home position.

SI Option Combi Dockable Table: Checking the emergency undock function

- Pull the emergency undocking handle (→ 3/Fig. 104 Page 98).

Fig. 104: Emergency undock handle



- » The docking station releases the Dockable Table. The table is undocked.
- Move the Dockable Table away from the docking station.
- Move the Dockable Table into the docking station.
- Step on the undocking pedal.
 - » The emergency undock function will be repositioned.
- Step on the docking pedal.
 - » The Dockable Table is docked.
- Step on the docking pedal again.
 - » Stepping on the pedals has no effect on the table.

4.5.8 Option Combi Dockable Table: Chassis and Wheels

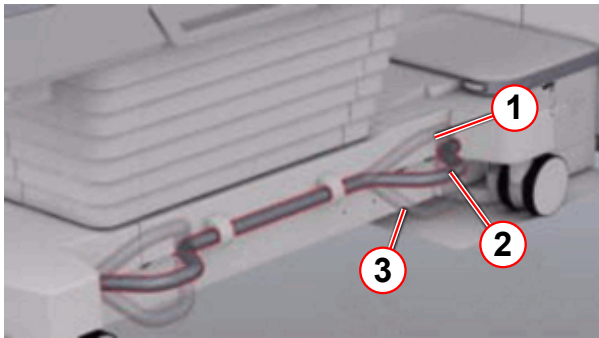
Required time: 5 min

Required tools: n.a.

SI Option Combi Dockable Table: Checking the chassis and the wheels

- Undock the Combi Dockable Table from the docking station.
- Move the table by a few meters.
 - » The table must move smoothly without any noticeable noise.

Fig. 105: Combi Dockable Table - lateral lever



- (1) 5th wheel is disabled
- (2) 5th wheel is activated
- (3) Table is braked

- Activate the 5th wheel by using the lateral lever (→ 2/Fig. 105 Page 99).
 - » The 5th wheel is activated.
- Move the table.
 - » The maneuverability is improved.
- Press the lateral lever (→ 3/Fig. 105 Page 99).
 - » It must not be possible to move the Combi Dockable Table

Table top lock

- Move the table top in a horizontal direction.
 - » It is possible to move the table top approximately 10 mm. Then stop the table top.

4.5.9 Option Combi Dockable Table: Lowering level of the undocked table

Required time: 5 min

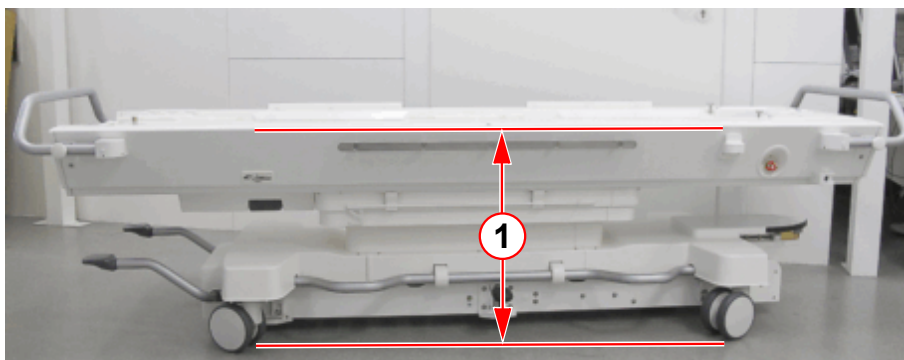
Required tools: n.a.

Prerequisites: The table is undocked.

SI Option Combi Dockable Table: Checking the lowering level of the undocked table

- Lower the patient table all the way by stepping on the down pedal.
- Check the lowering level
 - » The distance between the tabletop plate and the floor must be 600 mm or greater (→ 1/Fig. 106 Page 100).

Fig. 106: Combi Dockable Table



(1) Distance between tabletop and floor

4.5.10 Option Combi Dockable Table: Checking the interface box

Required time: 5 min

Required tools: n.a.

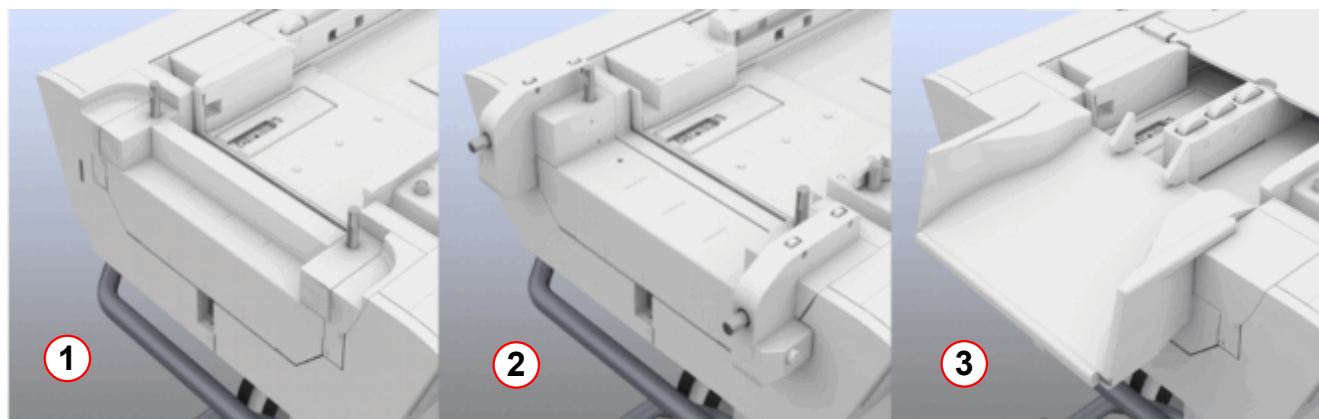
Prerequisites: The table is undocked.

SI Option Combi Dockable Table: Checking the interface box



Three types of the interface box are available depending on the different tables and transfer boards used for transfer to these tables:

Fig. 107: Combi Dockable Table

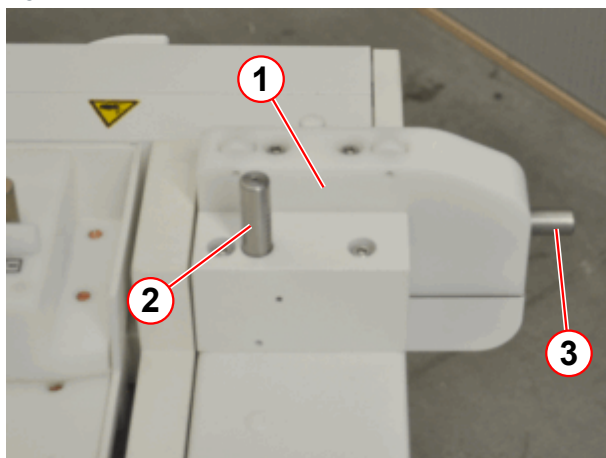


- (1) TRUMPF interface box
- (2) MAQUET interface box
- (3) Artis interface box

Test MAQUET interface box.

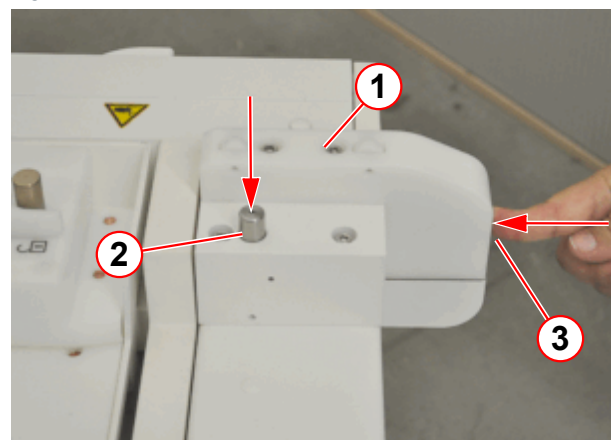
- Press the docking bolt (→ 3/ Fig. 109 Page 101).

Fig. 108: MAQUET Combi Dockable Table



- (1) MAQUET interface box
- (2) Locking bolt of the transfer box
- (3) Docking bolt

Fig. 109: MAQUET Combi Dockable Table



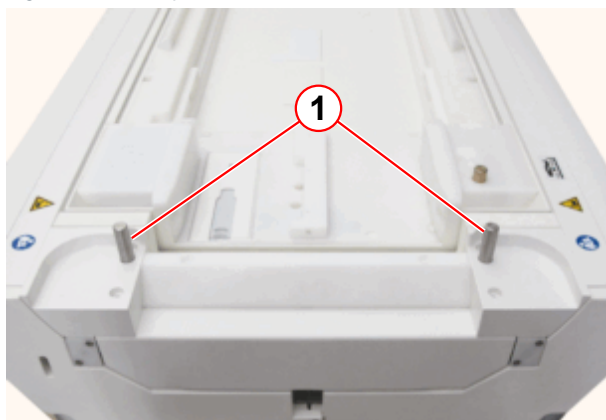
- (1) MAQUET Interface box
- (2) Locking bolt of the transfer box
- (3) Docking bolt

- » The locking bolt of the transfer board moves down (→ 2/ Fig. 109 Page 101).
- Release the docking bolt.
 - » The locking bolt of the transfer board moves up by spring force.
- Perform the test on the other locking bolt too.

Test the TRUMPF interface box.

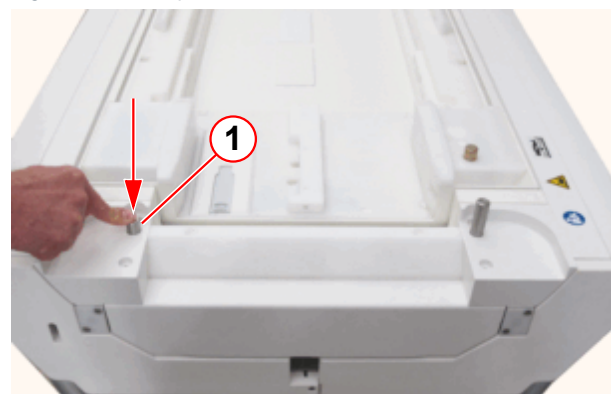
- Press the locking bolt (→ 1/ Fig. 111 Page 101).

Fig. 110: Trumpf interface box



- (1) Locking bolts of the interface box

Fig. 111: Trumpf interface box



- (1) Locking bolt of the transfer box is pressed

- Release the locking bolt.
 - » The locking bolt moves up by spring force.
- Perform the test on the other locking bolt too.

Test the Artis interface box

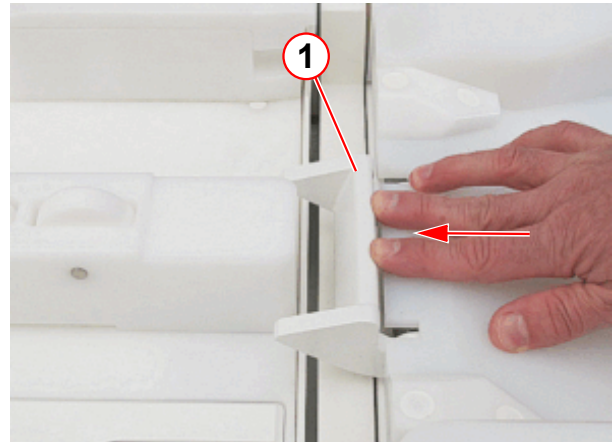
- Slide back the runaway stops (→ 1/ Fig. 113 Page 102)

Fig. 112: Artis interface box



(1) Runaway stops

Fig. 113: Artis interface box



(1) Runaway stops

- Release runaway stops
 - » The runaway stops moves back in starting position by spring force

4.5.11 Option Combi Dockable Table: Floor Dock

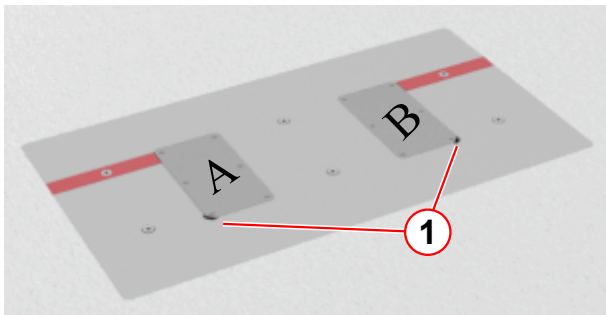
The floor dock is optional for the Combi Dockable Table.

Required time: 15 min

Required tools: n.a.

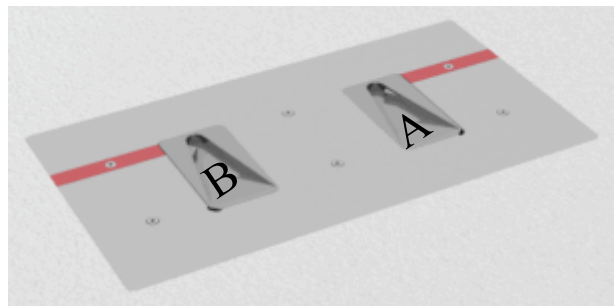
Prerequisites: The table is undocked.

Fig. 114: Floor dock - recesses closed



(1) Recess lifting aid

Fig. 115: Floor dock - recesses open



If the recesses are closed, insert a screw driver into the recess lifting aid (→ 1/ Fig. 114 Page 102) and lift the recess module.

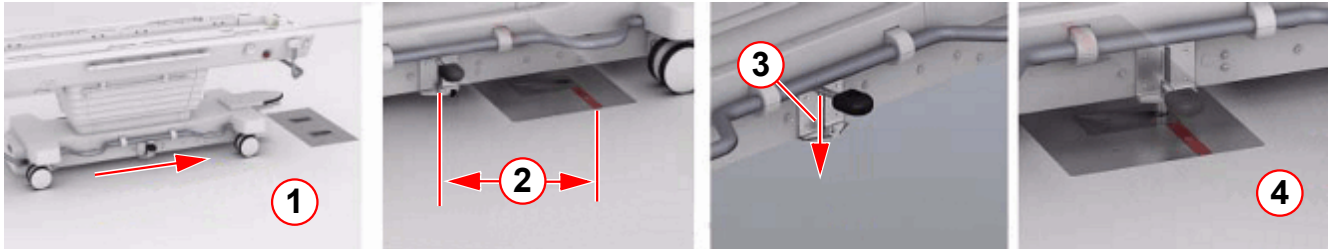
Turn the recess modules 180° and insert it crosswise.

SI Option Combi Dockable Table: Checking the floor dock

Perform a floor dock

- Move the Combi Dockable Table straight to the recesses on the floor and align it (→ 1/Fig. 116 Page 103).

Fig. 116: Floor dock



(1) Moving Combi Dockable Table

(2) Approximately 50 cm in front of the recesses

(3) Floor dock pedal

(4) Combi Dockable Table is docked

- If the table is positioned approximately 50 cm in front of the floor recesses (→ 2/Fig. 116 Page 103), slowly step on the pedal for floor dock (→ 3/Fig. 116 Page 103).
 - » Under the Combi Dockable Table, two bolts are pressed down via spring force.
- Slowly move the Combi Dockable Table completely into the funnel-shaped recesses until the pedals for floor dock are above the red marking line on the floor.
- Check that the two bolts are pressed down and fixed in the recesses on the floor (→ 4/Fig. 116 Page 103).
 - » The bolts are pressed down and fixed in the recesses on the floor.

4.6 Quality Assurance (QA)

Required time: 70 - 110 min depending on installed options.

Required tools: QA Phantom Setup

The quality measurement of the body coil is a fully automated procedure. The results are entered in the on-site database.



Use the Online Help to obtain detailed information regarding, e.g., table control, phantom positioning, or specific items and procedures. Press the 'Help' button in the header bar of the corresponding platform.

Preliminary steps

Starting the service platform

All actions of the quality assurance procedure are performed via the service platform.

- Start the Service platform > **Options** > **Service** > **Local Service**.

SI Quality assurance measurement

Go to the Home menu and press the corresponding button to access the **Quality Assurance** sub platforms.

- Start the measurement by clicking the "Start" button.
 - » The **Quality Assurance** check is successfully completed when every individual QA step indicates the status < **done** >.



If the results are out of tolerance, perform "Tune-Up" or troubleshooting, if necessary. Repeat "Quality Assurance".

4.7 End of Function Tests

End of section.

Initial Version

These instruction and the (→ M7-010.833.05 / Safety-Related Tests according to IEC 62353 / Page 1) replace the following document:

- Maintenance Instructions Combined Workflow

The Maintenance Instructions Combined Workflow was split in two documents:

- The maintenance instruction (→ Preventive Maintenance / M7-010.831.05)
- Safety- related test (these document) (→ M7-010.833.05 / Safety-Related Tests according to IEC 62353 / Page 1)

Chapter	Changes
All	The procedures in these instructions are taken over from the document 'Maintenance Instructions Combined Workflow' with minor changes. ■ RF room door: Checking the door lock of the RF room, moved from PM to SRT ■ Phantoms: Checking the phantoms, moved PM to SRT ■ Patient fan: Checking the air filter of the patient fan, moved PM to SRT

Version 02 compared to Version 01

Chapter	Changes
General Information	(→ Time Frame for the Safety-Related Tests / Page 9) re-worked

Version 03 compared to Version 02

Chapter	Changes
K02	Identification of the controlled access area: (→ Identification of the controlled access area (0.5 mT area) / Page 33) ■ Editorial changes

There are no Hazard IDs in this document.

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

healthcare.siemens.com/services

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

Print No.: M7-010.833.05.03.02 | Replaces: M7-010.833.05.02.02
Doc. Gen. Date: 03.16 | Language: English
© Siemens Healthcare GmbH, 2015

siemens.com/healthineers